Module Business Intelligence

Module code: CST3340

Coursework 2

Submission Date: 15/04/2022

Module Leader: Stelios Kapetenaki (Hendon)

Module Coordinator: Geethu Joy

Term (AY) 2021-22

School of Science & Technology

Department of Computer Engineering and Informatics

Table of Contents

Introduction 3
Data Analysis and Visualization 5
Selection of Data Mining Algorithm 16
Data Pre-processing 17
Data Mining 21
Data Ethics 26
Conclusion 28
References 31

Table of Tables : Table 1. Data attributes

Table of Figures:

Figure 1Heart 3
Figure 2 dataset 6
Figure 3 Total No. of people with heart disease and without heart disease 7
Figure 4 Calculated field 7
Figure 5. Total record 8
Figure 6 creating group and parameter 8
Figure 7 . Top number of chest pain in male and female 9
Figure 8. Top N filter 10
Figure 9 Top N filter 11
Figure 10. Top N filter 11
Figure 11 Exercise vs Disease 13
Figure 12 Exercise vs Disease using sorting 13
Figure 13Figure 13. Disease based on Chol, BP, and Thalassemia 14
Figure 14 male prone to have Disease 15
Figure 15 Does BP depends on Chol 16
Figure 16 Relationship between Thalassemia, CP, B_sugar, Vessel, and Chol 16
Figure 17Relationship between Thalassemia, CP, B_sugar, Vessel, and Chol (finding A) 17
Figure 18. Relationship between Thalassemia, CP, B_sugar, Vessel, and Chol(finding B) 17
Figure 19F. Correlation between Heart disease, Chol, and B.P.(A) 18
Figure 20Correlation between Heart disease, Chol, and B.P.(B) 18
Figure 21 Presence of heart disease 19
Figure 22 Decision Tree 19
Figure 23 Data pre processing (A) 22
Figure 24. Data pre processing (B) 23
Figure 25Data pre processing (c) 24
Figure 26 Dataset categorical 24
Figure 27. Visualization of all attributes in Weka 25
Figure 28 data mining process 26
Figure 29. data mining pre processing in weka 27
Figure 30 classifying j48 in weka 27

Figure 31classifier output in weka 28
Figure 32. confusion Metrix in weka 29
Figure 33. rules in weka 30
Figure 34 Decision tree in weka 31
Figure 35 Data Ethics 34

Introduction

Heart disease has become the leading cause of death. The development of computational algorithms that can reliably diagnose cardiac disease will reduce mortality rates and save money on healthcare.

Figure 1Heart

Typically, data analysis results are based on a restricted collection of variables from observational research. Various technologies can tell whether a patient has heart disease by analyzing vast data from various healthcare databases.

As a result, cardiology is ideal for inventing and employing computer methods to foresee cardiac problems and promising recent efforts to construct computer models that assess and predict heart illness.

This study aimed to develop an analysis of heart disease prediction whether a person has it or not.

Data Description

We used data sets gathered from four different institutions: the Cleveland Clinical Foundation in Cleveland, the Hungarian Institute of Cardiology in Budapest and the V.A Medical Center in Long Beach (Switzerland).

Medical test results and heart disease diagnoses for 270 people are included in the database. There were stages to the project. Data preprocessing and investigation were the primary goals of Phase I, which is documented in this paper. We will see more model construction when this project moves to its second phase. Data preprocessing is covered in Section 3, and each feature is analyzed in Section 4 to see how they relate to one another. In the concluding part, the findings are summarized.

The present dataset was taken from the **Statlog (Heart) Data Set from Kaggle.**

The dataset includes the details of attributes and variables. 14 attributes (including the target feature) and 270 instances were used in these analyses. The 'Class' field referred to the presence or absence of heart disease in patients and comprised an integer value for 0 (No Disease) to 1 (disease present). In this study, classifiers were built using the combined datasets, and their performance was evaluated using cross-validation. Diagnosis of heart disease (angiographic disease status). It was designated as diameter narrowing in any major vessel.

The following is the variable description extracted from the Statlog (Heart) Data Set file:

Number	Attributes	Description	Datatype	Domain
1	Age	Patient age in year	Numerical	29 to 77 years
2	Sex	Gender	Binary	0=Female, 1=Male
3	CP	Chest Pain	Categorical	1=typical angina 2=atypical angina 3= nontypical angina 4 = asymptomatic angina
4	BP	Resting blood pressure	Numerical	94 to 200 Categorized in ≥120 and >120
5	Chol	Serum Cholesterol	Numerical	116 to 409 Normal = <220 High = 220 – 329 High Risk = >240
6	Bsugar	Fasting Blood Sugar	Binary	0 = Normal 1 – High
7	ECG	Resting electrocardiographic result	Categorical	0 = Normal 1 = having ST-T wave abnormality 2 = Left ventricular hypertrophy
8	H_rate	Maximum heart rate	Numerical	71 to 200 Categorized in Normal and High
9	Ex_angina	Exercise include angina	Binary	0 = No 1 = yes
10	Oldpeak	Old peak	Numerical	Continuous (0 to 6.2 Categorized as: <1.3 = low ≥1.3 = High
11	Slope	Slope of peak exercise ST segment	Categorical	1 = upsloping 2 = Flat 3 = downsloping
12	Vessel	Number of major vessels	Categorical	0 = Not coloured 1,2,3 = coloured
13	Thalassemia	Defect Type	Categorical	3 = Normal

Data Analysis and Visualization

In this section, exploratory data analysis (EDA) is carried out to find the hidden patterns in the chosen dataset (Heart statlog Dataset). Tableau is the software used to create a visualization of the dataset.

Exploratory Data Analysis:

1. Missing value: 0
2. Outlier: 0
3. No. of people with heart Disease: 120
4. No. of people without heart Disease: 150
5. Variable to be predicted: Absent (0) or present (1)

Age	sex	CP	BP	Chol	B_sugar	ECG	H_Rate	Ex.Angine	Oldpeak	Slope	Vessele	Thalassemia
U80	Male	asymptomaticangina	High	High Rsk	High	Abnormal	Normal	Yes	High	Flat	Coloured	No
U70	Female	nonangina	Normal	High Rsk	Low	Normal	High	No	Low	Flat	Not Coloured	Fixed
U60	Male	atypical angina	High	High Rsk	High	Abnormal	High	Yes	Low	Upsloping	Not Coloured	Reversible
U70	Male	nonangina	High	Normal	High	Normal	Normal	Yes	Low	Flat	Not Coloured	Fixed
U80	Female	nonangina	Normal	Normal	High	Hipertrophy	Normal	Yes	Low	Upsloping	Not Coloured	No
U70	Male	nonangina	Normal	Normal	High	Normal	High	No	Low	Upsloping	Not Coloured	No
U60	Male	asymptomaticangina	High	High Rsk	High	Abnormal	High	Yes	Low	Flat	Coloured	Fixed
U60	Male	asymptomaticangina	Normal	High	High	Abnormal	High	Yes	Low	Flat	Coloured	Reversible
U70	Male	asymptomaticangina	High	High Rsk	High	Abnormal	High	Yes	High	Flat	Coloured	Reversible
U70	Female	asymptomaticangina	High	High Rsk	High	Hipertrophy	High	No	High	Flat	Coloured	Reversible
U60	Male	nonangina	Normal	Normal	Low	Abnormal	High	No	Low	Flat	Not Coloured	No
U60	Male	nonangina	Normal	Normal	Low	Abnormal	Normal	Yes	Low	Upsloping	Not Coloured	No
U50	Male	nonangina	Normal	High	Low	Abnormal	High	No	Low	Upsloping	Not Coloured	No
U70	Male	typical angina	High	High	Low	Normal	High	No	High	Flat	Coloured	Fixed
U60	Female	nonangina	High	Normal	Low	Hipertrophy	High	No	Low	Upsloping	Not Coloured	No
U80	Female	nonangina	Normal	Normal	Low	Normal	Normal	No	Low	Upsloping	Not Coloured	No
U50	Male	asymptomaticangina	High	High Rsk	High	Abnormal	Normal	Yes	High	Flat	Coloured	Reversible
U60	Male	asymptomaticangina	High	High Rsk	High	Abnormal	High	Yes	High	Downsloping	Coloured	Reversible
U70	Male	typical angina	Normal	High	Low	Normal	High	No	High	Upsloping	Not Coloured	No
U50	Male	nonangina	High	Normal	High	Normal	High	Yes	High	Upsloping	Not Coloured	Reversible
U70	Male	asymptomaticangina	High	High	High	Abnormal	Normal	Yes	High	Flat	Coloured	Reversible
U50	Male	nonangina	Normal	High	Low	Normal	High	No	Low	Upsloping	Not Coloured	No

Figure 2 dataset

Initial analysis of the data using visualization techniques within Tableau

In the first visualization, a "Percent of total" table calculation was applied to get the percentage distribution of heart disease records present or absent in patients. A calculated field called Total Records was also created to find the total number of all records available. See below:

1. **The number of persons with heart disease vs non-heart disease:**

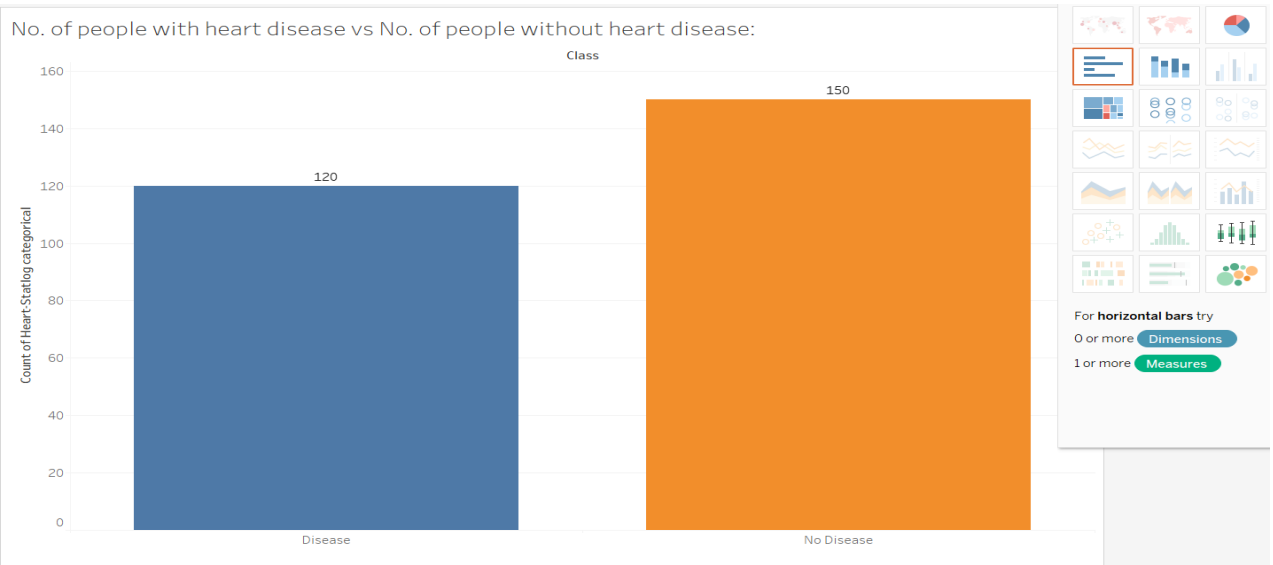


Figure 3 Total No. of people with heart disease and without heart diseas

Figure 3. Total No. of people with heart disease and without heart disease

We have 120 with heart problems and 150 without it. However, to get the total record in percentage, it is needed to use the calculated field application in Tableau. The results' present and absent' can be obtained using the "IF ELSE IF" logical operator.

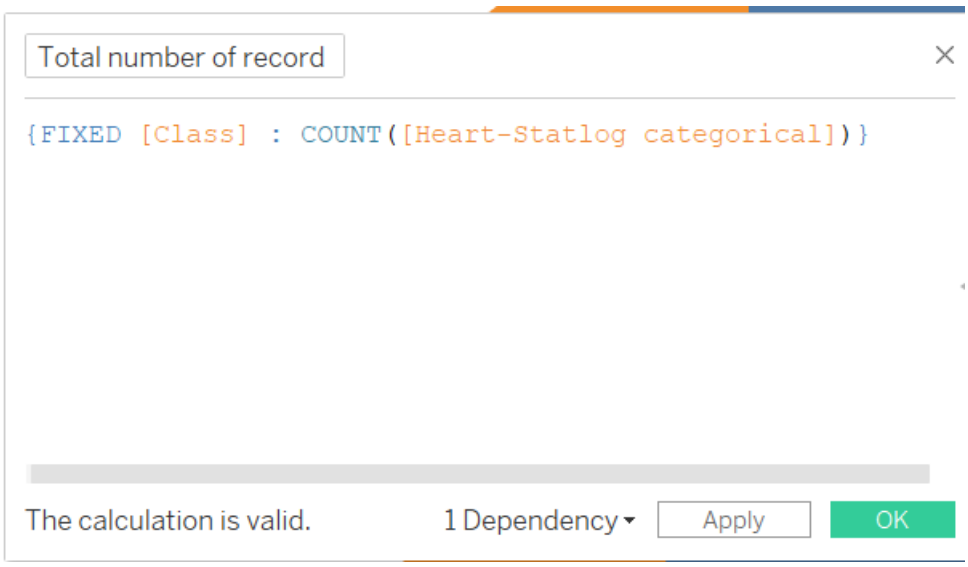


Figure 4 Calculated field

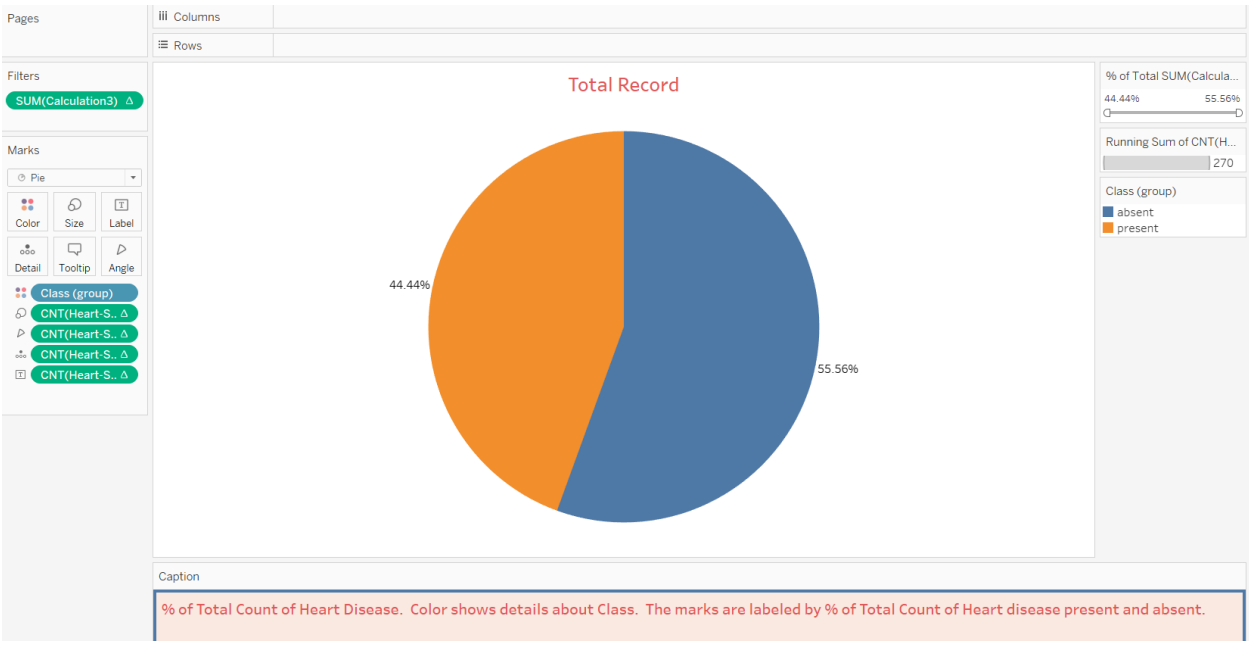


Figure 5. Total record

2. Top N Categories of Chest Pain in Male and Female

In this visualization, the top N category of Chest pain in Male and Female individuals is displayed using the Parameters and filter option by Tableau. Firstly, the Sex attribute is filtered by females to create the parameter as Top Female. Secondly, the male filters the same attribute to create a parameter as Top Male. The slider option is used in this visualization.

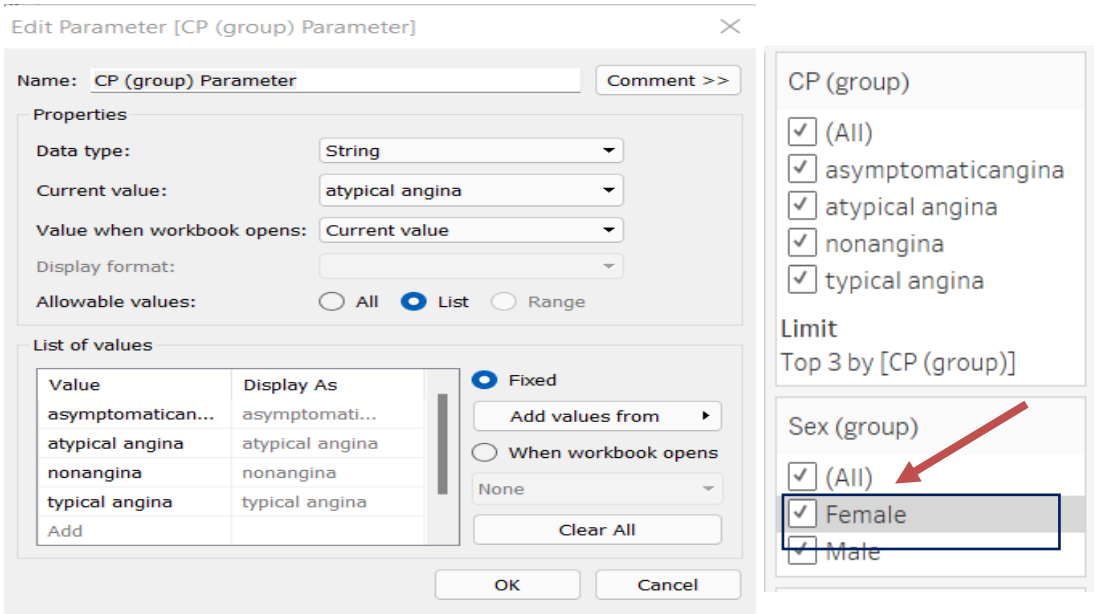


Figure 6 creating group and parameter

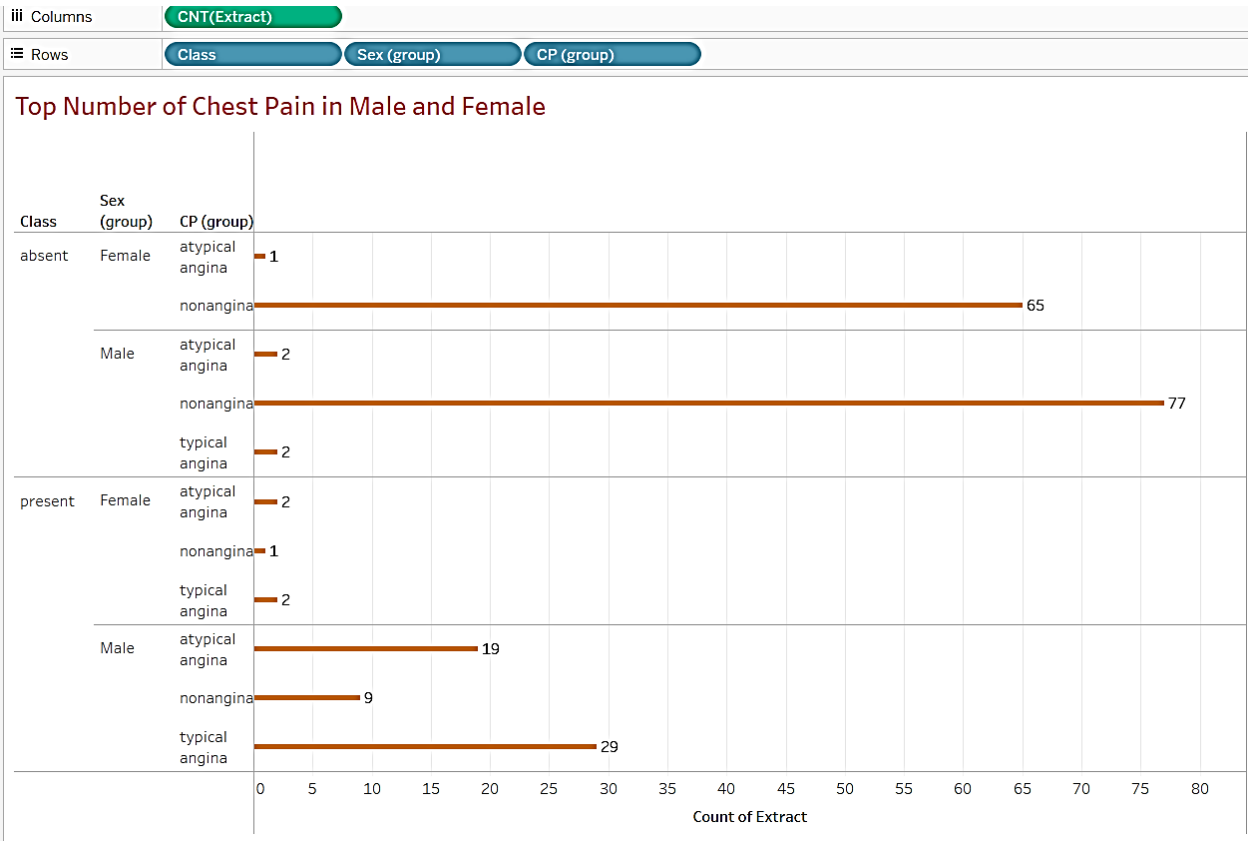


Figure 7 . Top number of chest pain in male and female

This graphic depicts four types of chest discomfort (typical angina, atypical angina, nontypical angina, asymptomatic angina). Top 3 Chest pain 3 people had atypical angina, but no cardiac disease. 2 females and 19 males had cardiac disease with atypical angina chest

pain. Only one Female and nine male had non-angina chest pain with heart disease. However, most males and females do not have heart disease and experience nonanginal discomfort.
Also, men had the most angina chest pain with heart disease, while females had just two. Looking at the

Similar parameters are utilized to identify the top number of Age groups with and without heart disease. Thus, the visualization shows just the top 5 age groups, each represented by a distinct color.

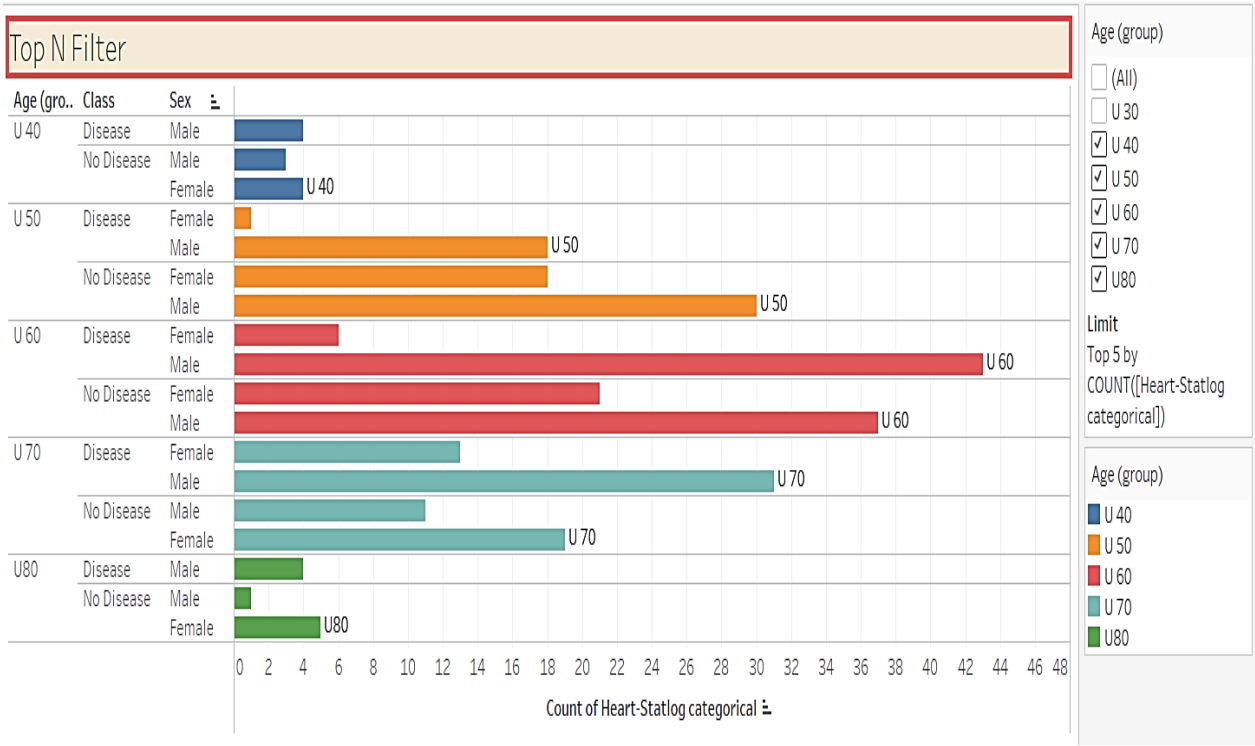


Figure 8. Top N filter

Figure 9 Top N filter

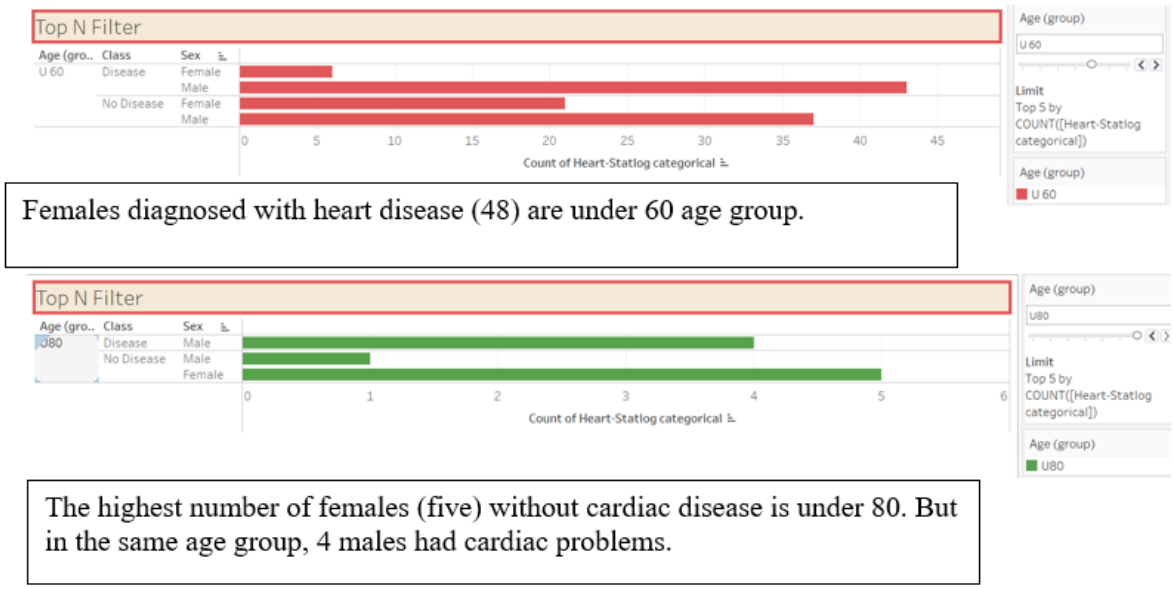


Figure 10. Top N filter

The most common age group for those with heart disease is 48.

3. Heart Disease compared with Exercise angina.

Exercise-induced angina is a severe pain that feels tight, gripping or squeezing.

Finding

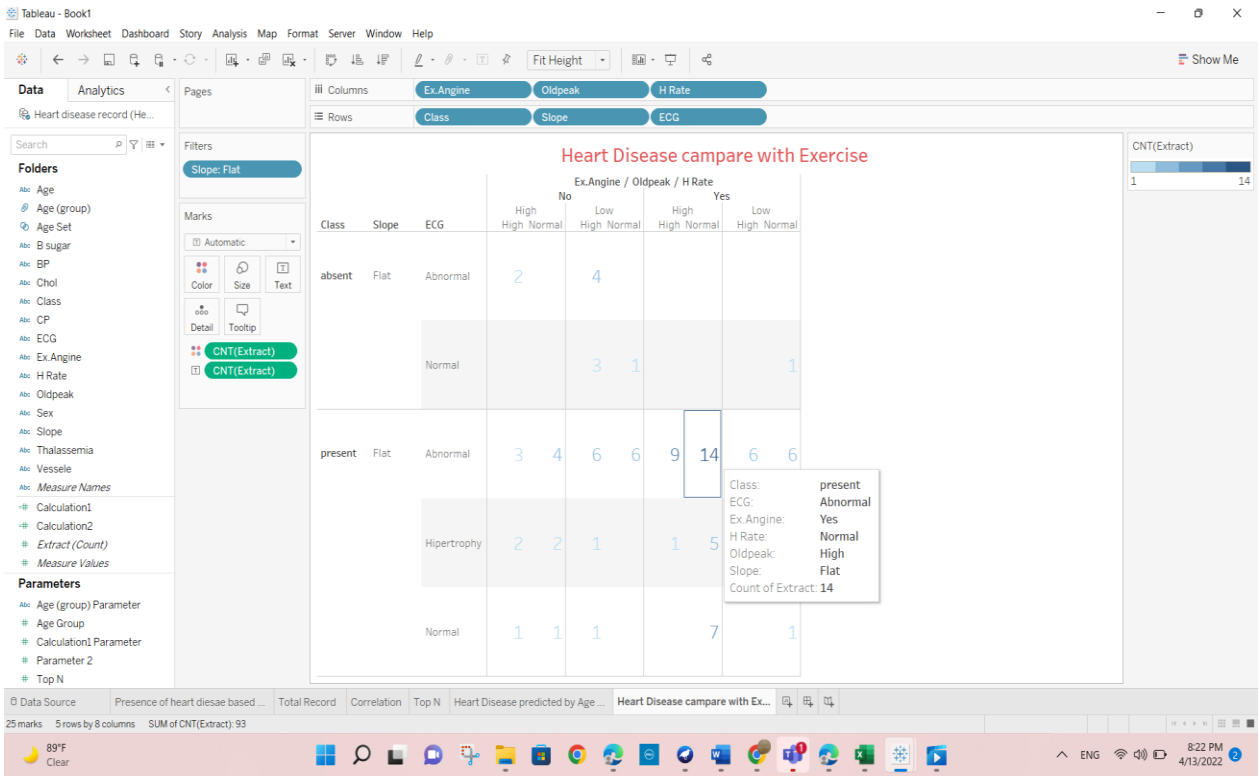


Figure 11. Exercise-induced angina vs Dis

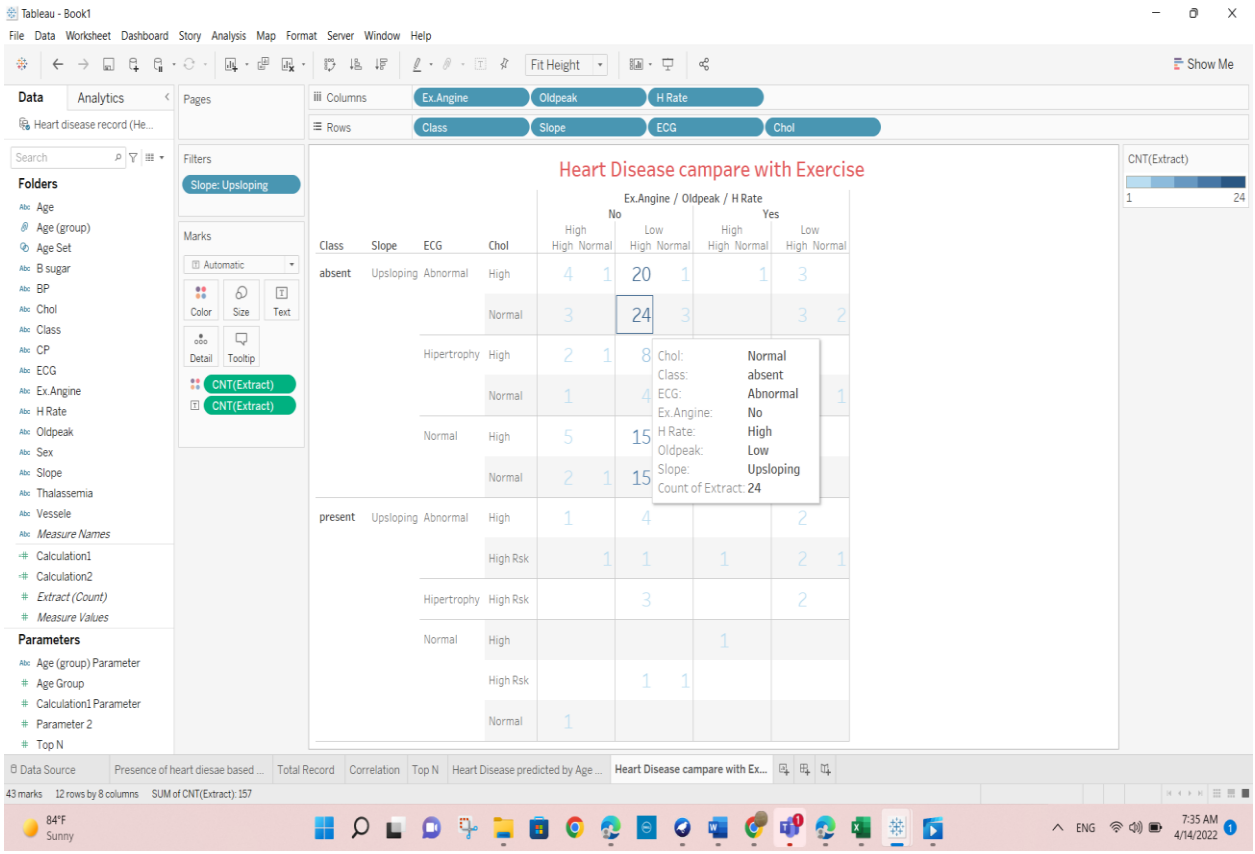


Figure 11 Exercise vs Disease

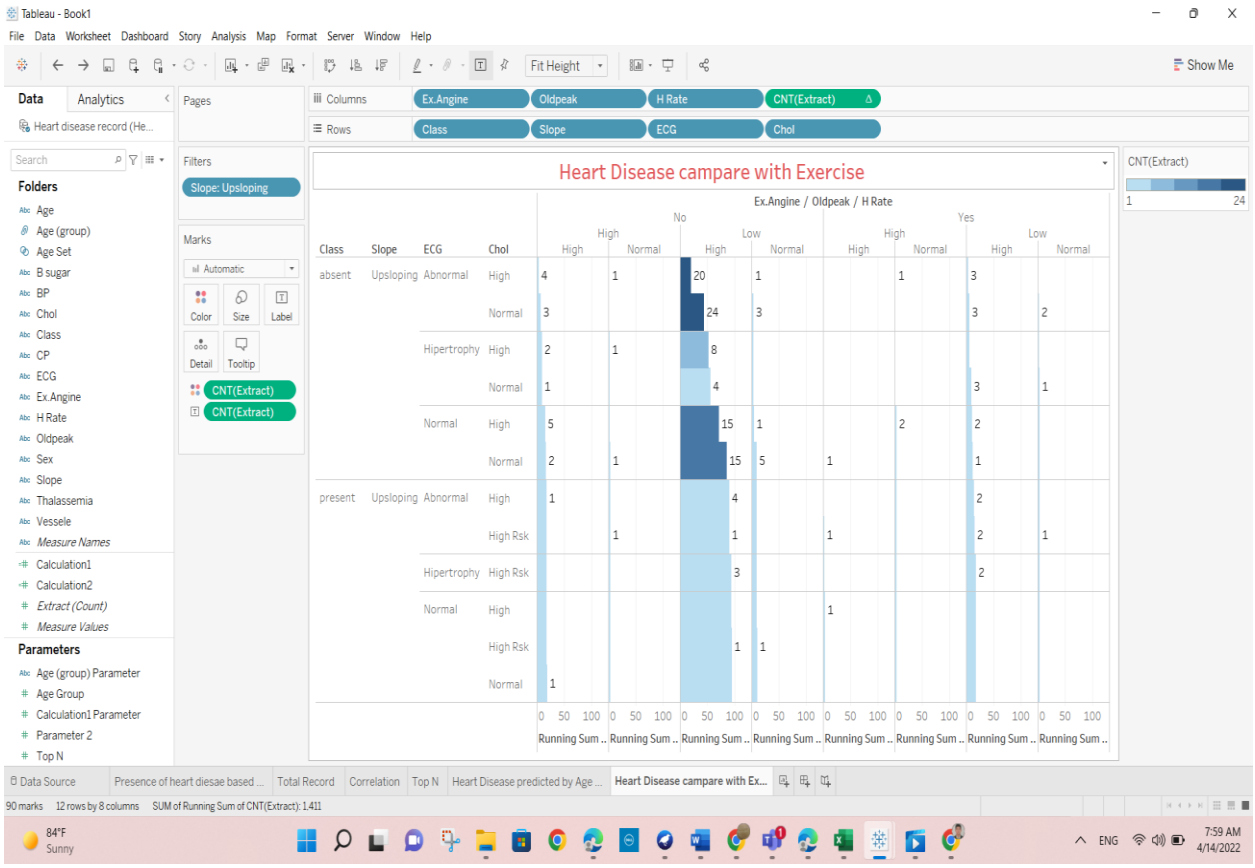


Figure 12 Exercise vs Disease using sorting

4. Disease present based on Chol, B.P., and Thalassemia

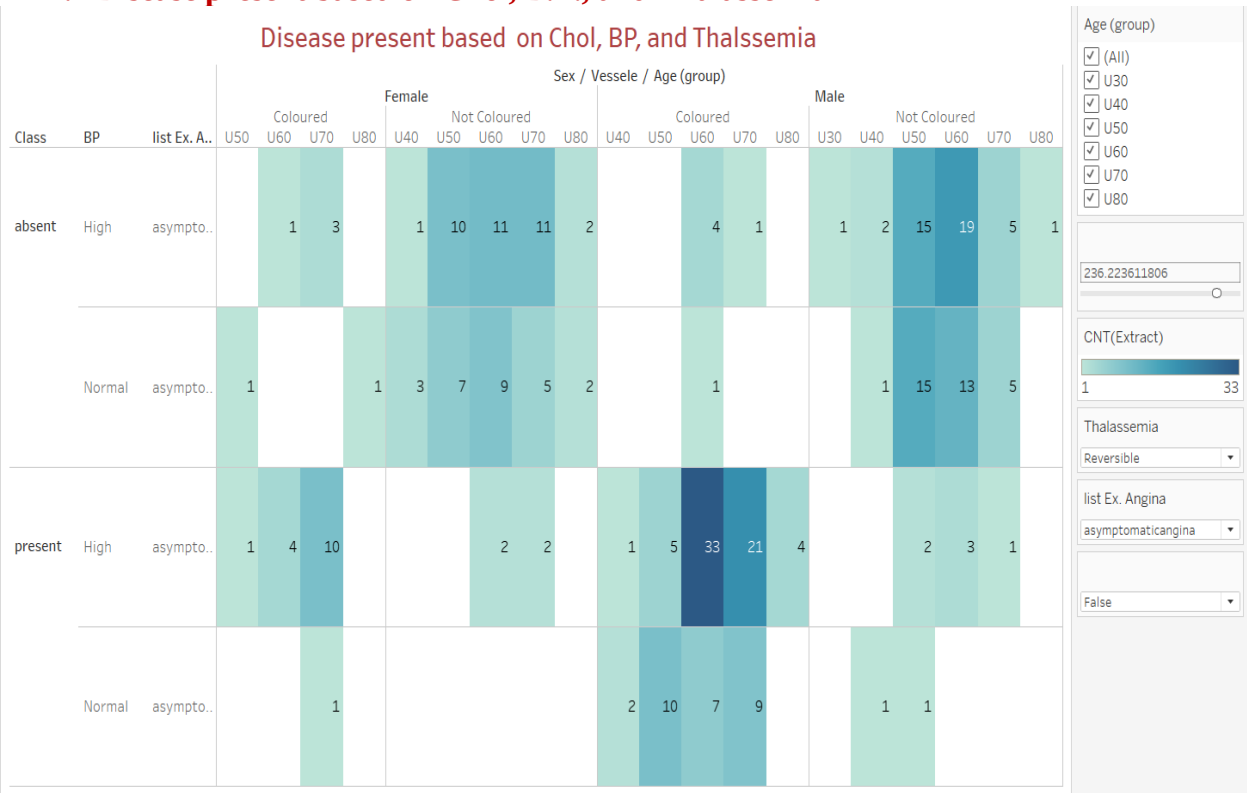


Figure 13Figure 13. Disease based on Chol, BP, and Thalassemia

5. Are male prone to have heart disease than Female?

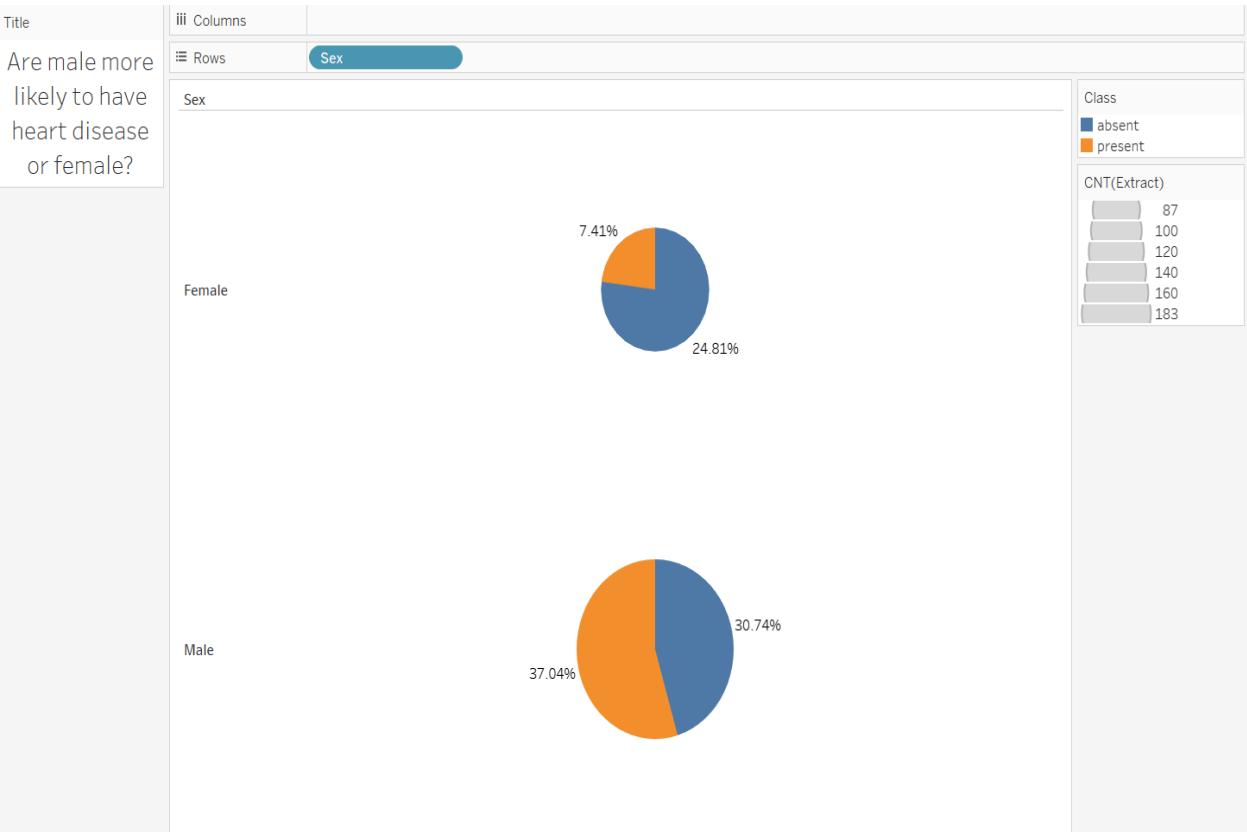


Figure 14 male prone to have Disease

The above visualization displays that 37% of males have heart disease, whereas females are only 7.4%. In this visualization, parameter and filter options are used.

6. Does Blood pressure depend on Cholesterol?

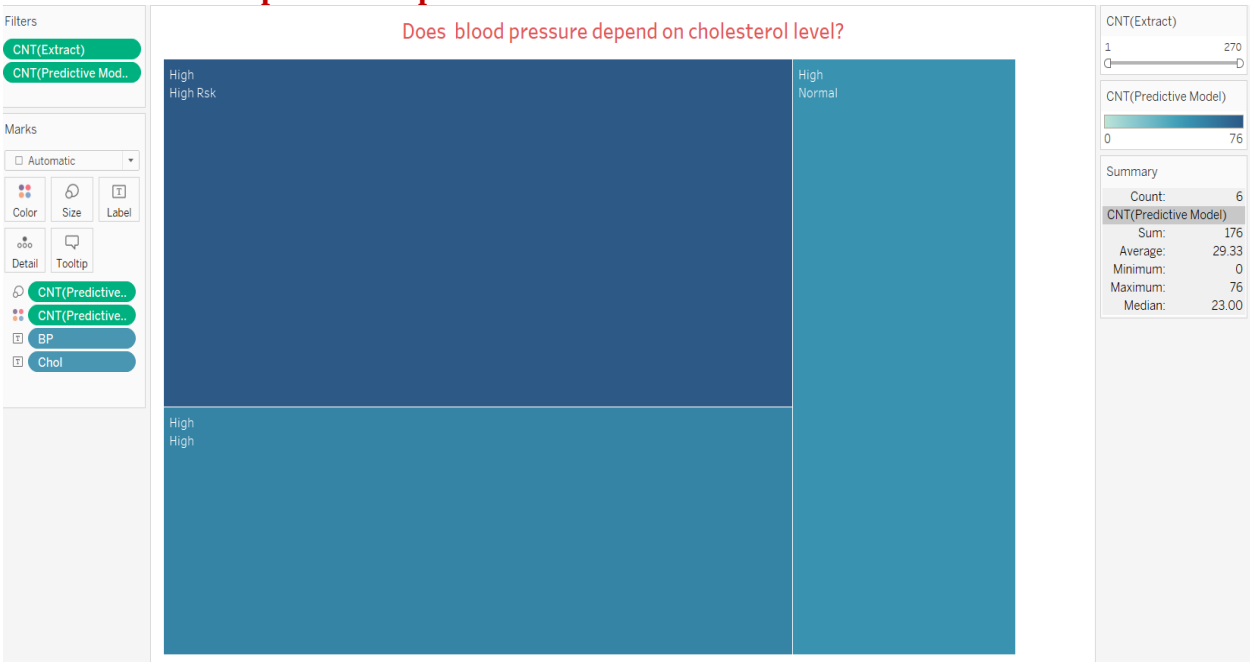


Figure 15 Does BP depends on Chol

The above visualization shows76 individuals have high B.P. and high-risk Cholesterol; 54 individuals have both B.P. and Chol are high. At the same time, only 46 individuals are different.

7. Is there a relationship between Thalassemia, CP, B_sugar, Vessel, and Chol?

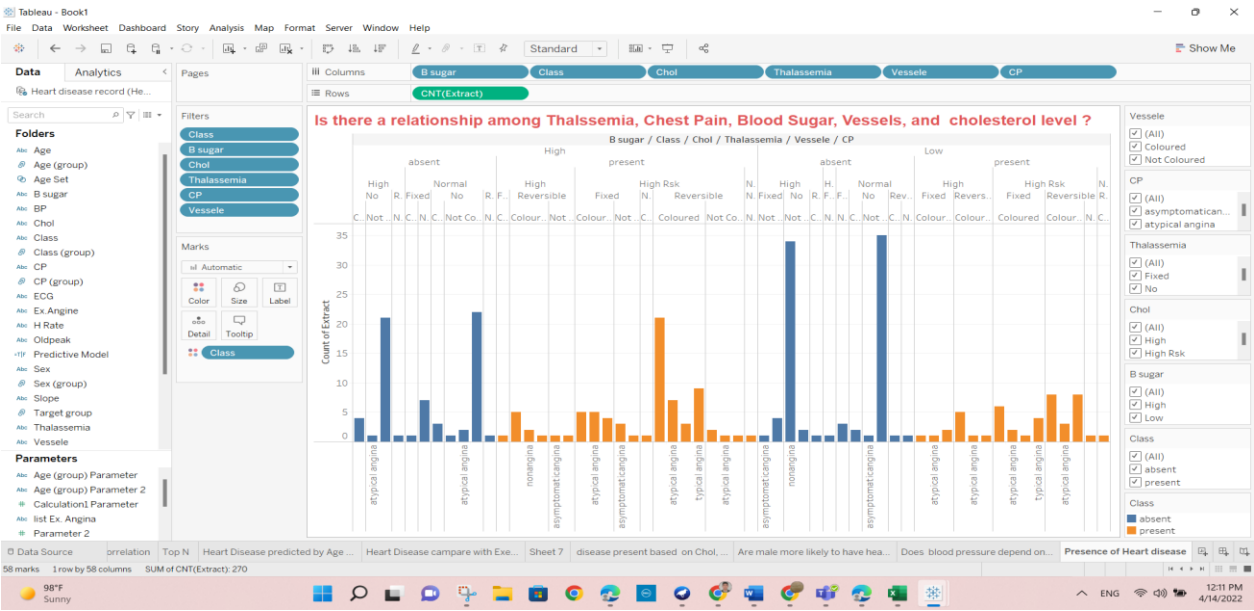


Figure 16 . Relationship between Thalassemia, CP, B_sugar, Vessel, and Chol

In the above output, used all attributes have a relationship among them. If Chol, B.P., Thala is high, the patient has the disease. Similarly, if any two variables are normal or low, patients

do not have the disease. The calculation field, functions filter, parameter, and sorting methods are used here.

Finding

A.

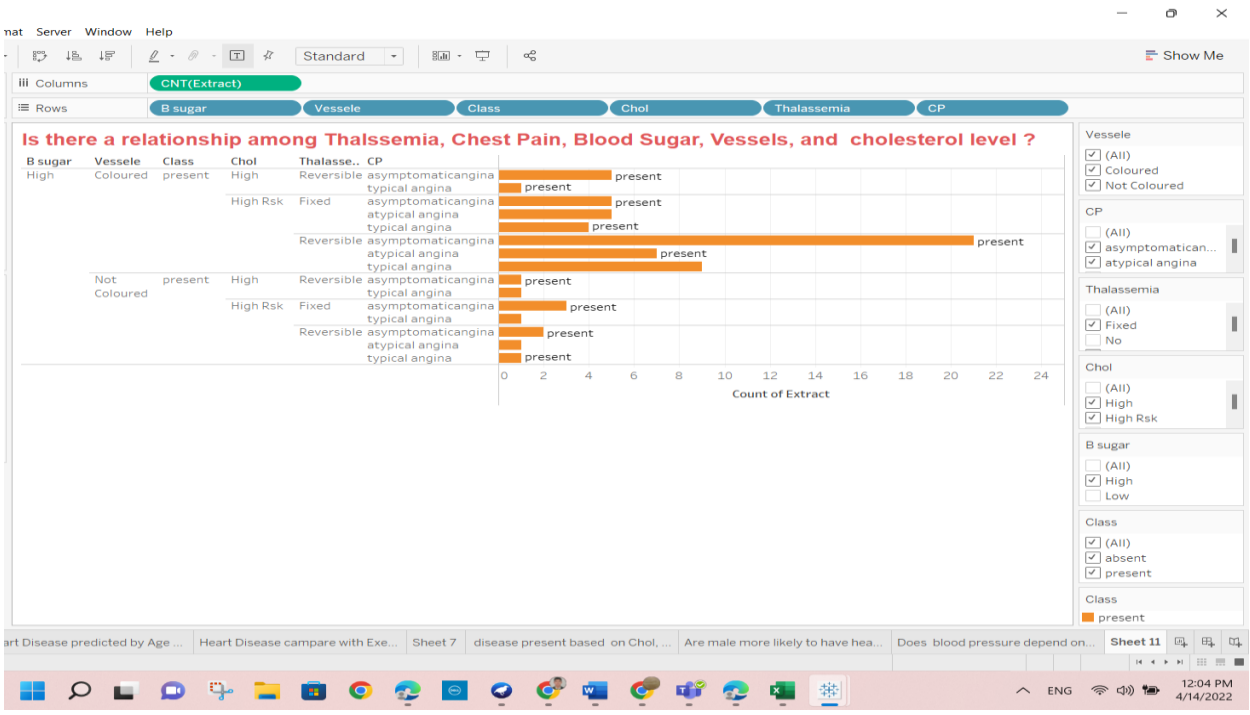


Figure 17Relationship between Thalassemia, CP, B_sugar, Vessel, and Chol (finding A)

B.

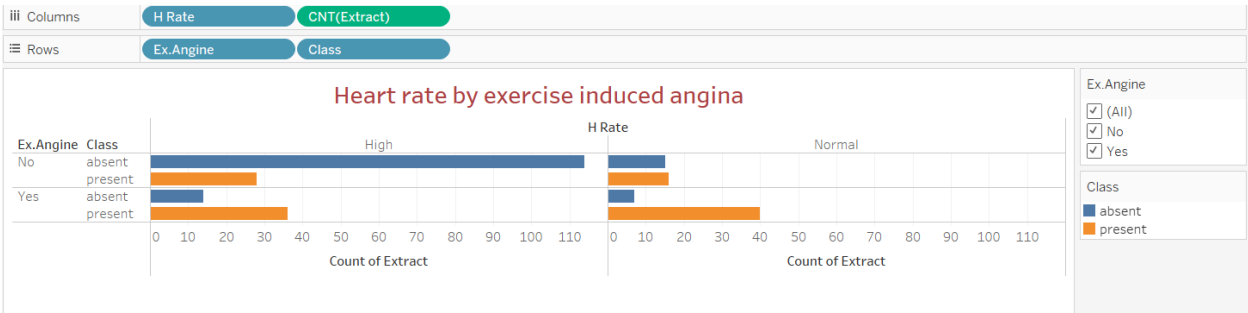


Figure 18. Relationship between Thalassemia, CP, B_sugar, Vessel, and Chol(finding B)

8. Correlation between Heart disease, Chol, and B.P.

This visualization shows the relationship between heart disease and Cholesterol levels. To get the result, analysis was made through the calculated field, a total number of male and Female, variables of chol and B.P. High B.P. with High Chol. Leads to Heart disease. A smaller number of patients did not have disease though their chol and B.P. were high. It may be some other factors are responsible.

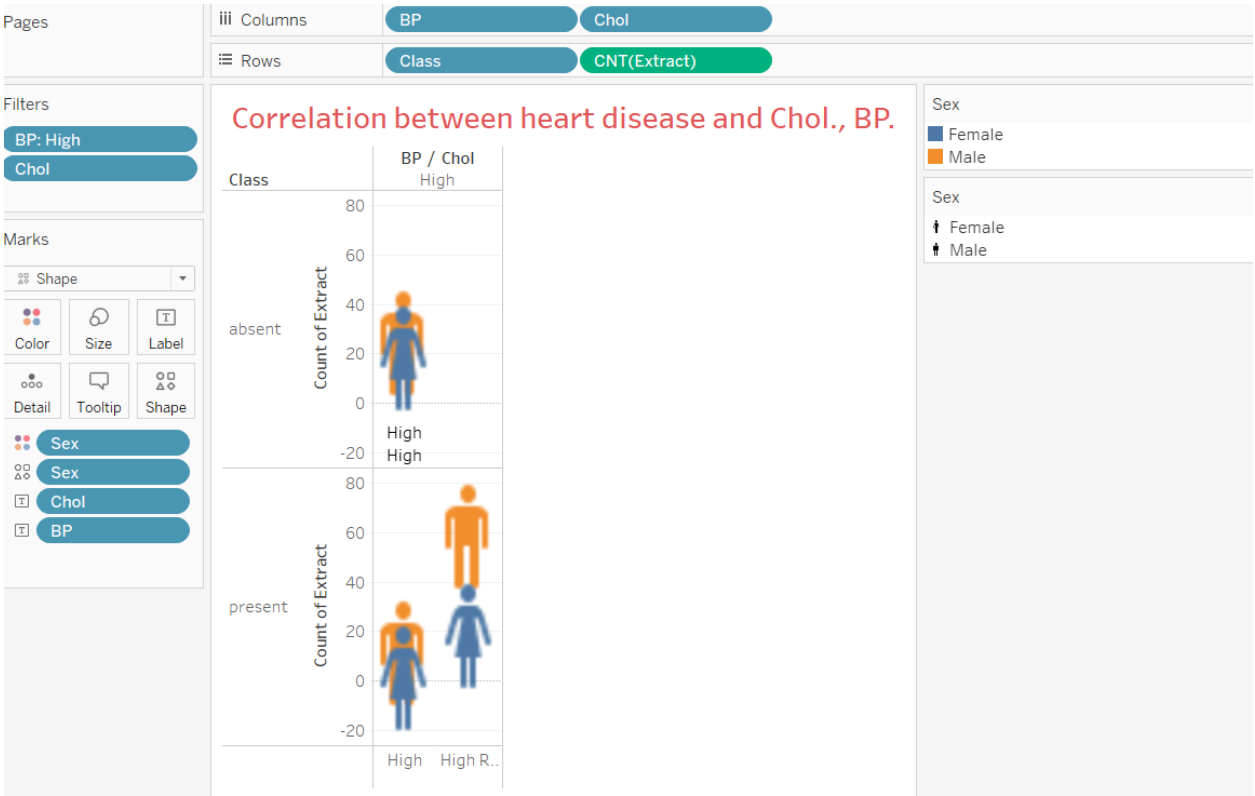


Figure 19Figure 18. Correlation between Heart disease, Chol, and B.P.(A)

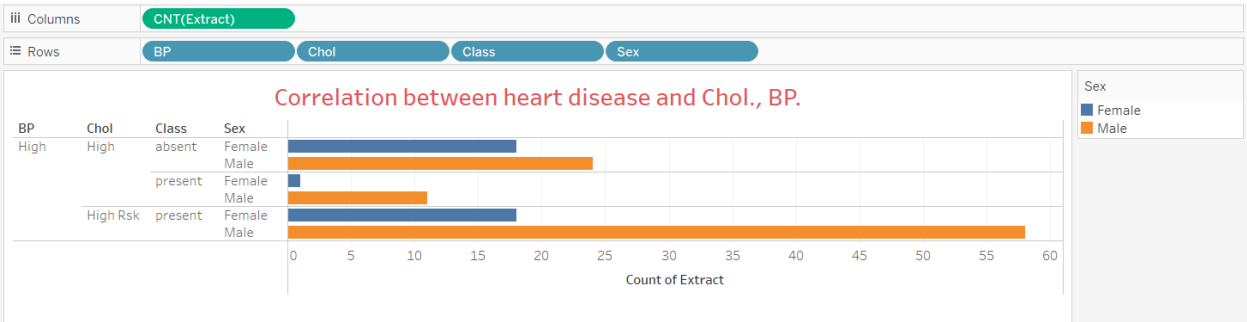


Figure 20Figure 19. Correlation between Heart disease, Chol, and B.P.(B)

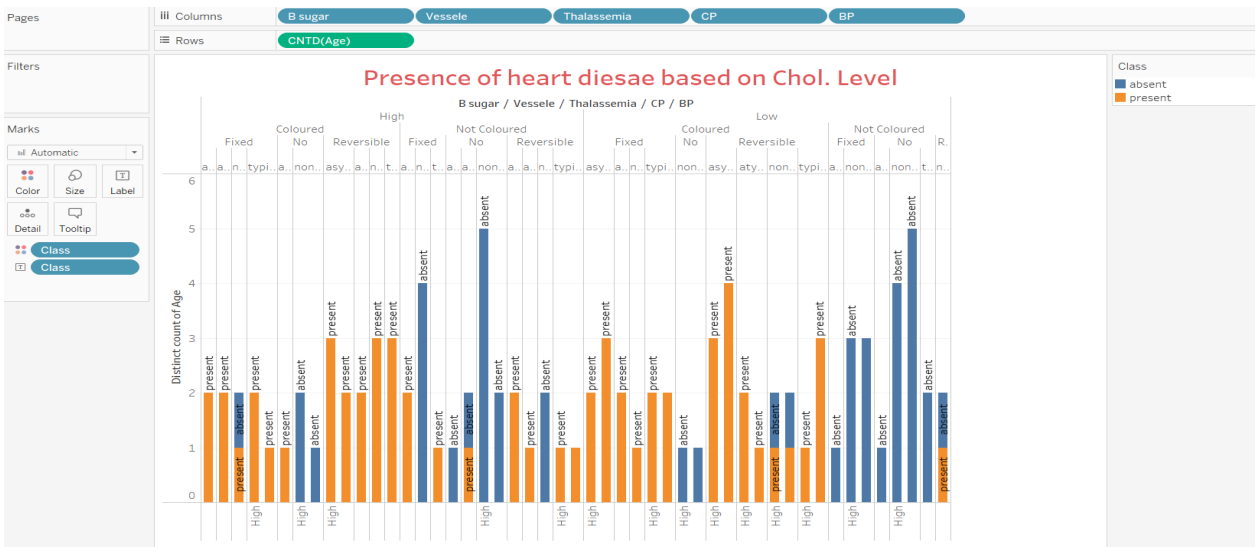


Figure 21Figure 20. Presence of heart disease

Selection of Data Mining Algorithm

As a result of the data analysis on Tableau, 'Decision Trees' has been chosen as the data mining algorithm.
A Decision Tree is a data mining approach used to develop Classification Models. It develops categorization models in a tree-like structure, as its name implies. This is guided class learning. (*Decision Tree Algorithm Examples in Data Mining*, n.d.)
Decision trees are used in two ways.
• **Classification Analysis – is used for** Categorical Variables where decision trees target categorical variables.
• **Regression Analysis – It is used for** Continuous Variables where decision trees are used for continuous variables.
A decision tree is a flowchart tree made up of training set tuples. The dataset is divided into smaller subgroups and shown as tree nodes. A tree has root nodes, internal nodes, leaf nodes, and branches.

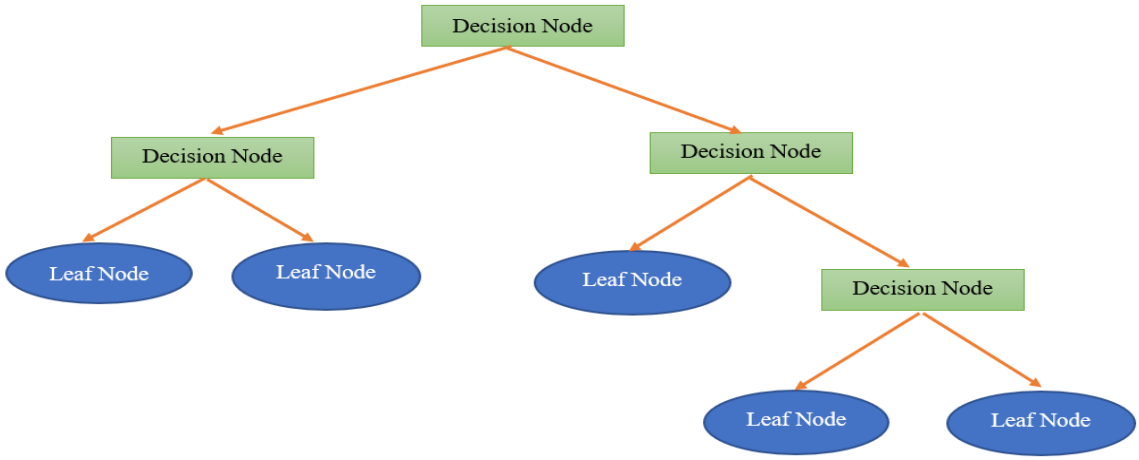
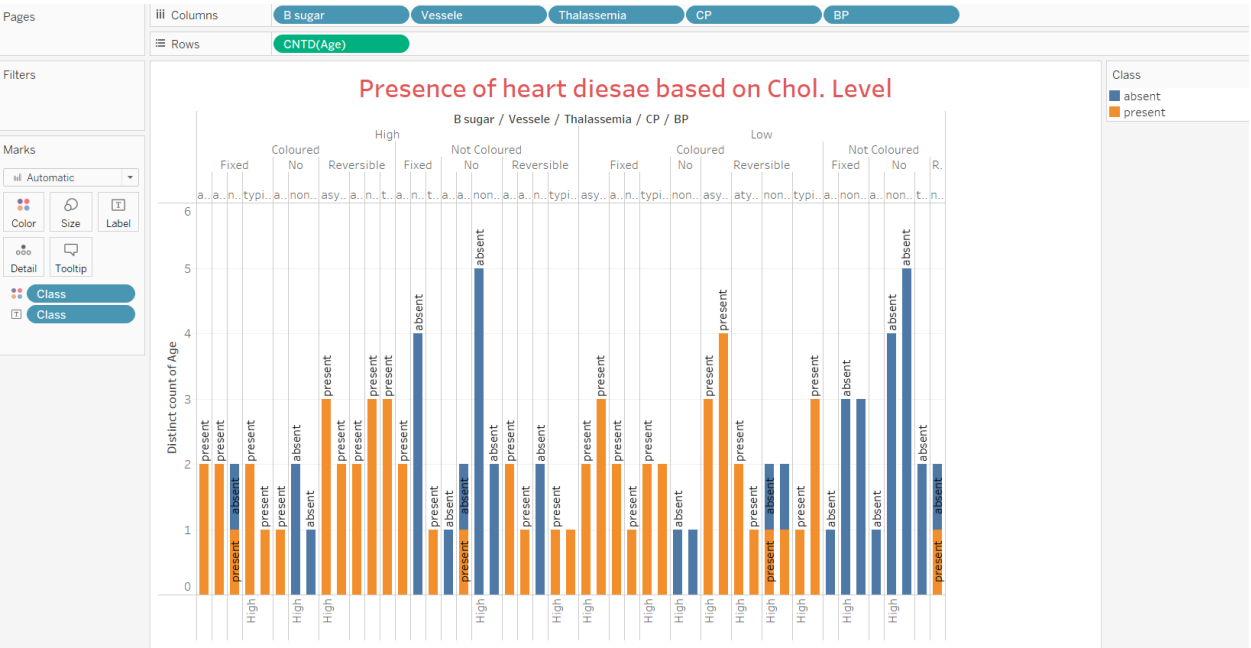


Figure 22Figure 21. Decision Tree

Justification

The decision tree approach was selected for this data set since it had an explicit class variable to forecast. The status of heart disease prediction act as a class variable, whether the individual is diagnosed with the presence of heart disease or absence.
Several attributes such as Blood pressure, Cholesterol, Blood sugar, Exercise angina, Thalassemia, old peak, Chest pain, and Vessels are considered to determine whether the patients
Have heart disease. A decision tree also delivers more precise and intuitive outcomes.
As shown below, the prediction of heart disease purely depends on the percentage
Of cholesterol level, chest pain intensity, type of Thalassemia, blood pressure level, and No. of heart vessels coloured by fluoroscopy. It is identified that as the

These variables are high; the patients are more likely to present heart disease. Thus, this is a classification dataset where you categorize the heart disease prediction ('absence' or "present").



In this visualization, the Patients status is based on Chol, CP, B.P., and Thalassemia attributes, classifying them based on their level achieved into the present (orange) & absent(blue). Since this is a classification dataset, we used the decision-tree-algorithm

Data Pre-processing

Data Pre-processing transforms raw data into a usable format. Data mining requires it since we cannot deal with raw data. Ensure the data is good before utilizing machine learning or data mining.

Identify your input and class variables, if relevant (i.e. which variable are you going to consider for your class variables).

The present dataset is based on the prediction of heart disease for patients. There are 270 instances with 13 attributes, along with the class variable. The measures in this dataset include the level of variables of the patients of age, sex, B.P. (blood pressure), B sugar(fasting blood sugar), Ex_angina (exercise-induced angina), C.P. (chest pain type), ECG (resting electrocardiographic results), H Rate (maximum heart rate achieved), old peak (S.T. depression induced by exercise relative to rest), Slope(the Slope of the peak exercise S.T. segment), Vessels (number of significant vessels (0-3) coloured fluoroscopy), and Thalassemia. The class variable of the present dataset is the status attributes that refer to the diagnosis of heart disease, present or absent.
The values for this dataset are either 'placed' or 'not placed.'
pending upon several factors. The aim of mining this dataset is to identify the factors that

usually present or are absent from disease.
The dataset had no missing values. The dataset also did not possess any outliers as the data majorly concentrated on the prediction of heart disease that was clear in terms of the levels diagnosed through the clinical test.
All the other values across 14 attributes were pure and accurate with non-relevant anomalies such as outliers.
The dataset majorly constituted nominal and continuous, and binary types of data. Nominal data included chest pain, ECG, Slope, vessel, and Chol.
The continuous data were demanded to be converted to categorical/ordinal type for applying the decision tree algorithm, which only processes into ordinal type. Therefore, all the numerical and binary data were converted into a categorical data type.

Age	Age	age
70	U80	29
67	U70	34
57	U60	34
64	U70	35
74	U80	35
65	U70	35
56	U60	37
59	U60	37
60	U70	38
63	U70	39
59	U60	39
53	U60	39
44	U50	40
61	U70	40
57	U60	40
71	U80	41
46	U50	41
53	U60	41
64	U70	41
40	U50	41
67	U70	41
48	U50	41
43	U50	41
47	U50	41
54	U60	42
48	U50	42
46	U50	42

Variable type	Integer		
Missing	None		
Freq Distribution			
Categories	value	freq	
U30	20-29	1	
U40	30-39	11	
U50	40-49	67	
U60	50-59	107	
U70	60-69	74	
U80	70-79	10	

Mean	**54.43**
St Dev	**9.11**
Min	**29**
Q1	**44**
Median	**55**
Q3	**54**
Max	**77**
Range	**48**
IQR	**10**
Mode	**54**

Figure 23Figure 22 Data pre processing (A)

Here, the Age variable is numerical was categorized in six categories U30 to U80 where age distribution was 29 to 77 years of age and U60 was considered high.

sex	sex		sex						
1	Male		0			Variable type	Categorial		
0	Female		0			Missing	None		
1	Male		0						
1	Male		0						
0	Female		0				Value	Freq	
1	Male		0				0 Female	87	
1	Male		0				1 Male	183	
1	Male		0						
1	Male		0						
0	Female		0		**Mean**	**0.68**			
1	Male		0		**St Dev**	**0.47**			
1	Male		0						
1	Male		0		**Min**	**0**			
1	Male		0		**Q1**	**0**			
0	Female		0		**Median**	**1**			
0	Female		0		**Q3**	**1**			
1	Male		0		**Max**	**1**			
1	Male		0						
1	Male		0		**Range**	**1**			
1	Male		0		**IQR**	**1**			
1	Male		0						
1	Male		0		**Mode**	**1**			
1	Male		0						
1	Male		0						
0	Female		0						
0	Female		0						
0	Female		0						

Figure 24Figure 23. Data pre processing (B)

This attribute was denoted as binary data categorized in male and Female, where the percentage of the male is high.

CP	CP		CP						
4	asymptomaticangina		1			Variable type	Categorial		
3	nonangina		1			Missing	None		
2	atypical angina		1						
3	nonangina		1						
3	nonangina		1				Value	Freq	
3	nonangina		1				1 typical angina	33	
4	asymptomaticangina		1				2 atypical angina	24	
4	asymptomaticangina		1				3 nonangina	152	
4	asymptomaticangina		1				4 asymptomaticangina	61	
4	asymptomaticangina		1						
3	nonangina		1						
3	nonangina		1		**Mean**	**2.89**			
3	nonangina		1		**St Dev**	**0.89**			
1	typical angina		1						
3	nonangina		1		**Min**	**1**			
3	nonangina		1		**Q1**	**2**			
4	asymptomaticangina		1		**Median**	**3**			
4	asymptomaticangina		1		**Q3**	**3**			
1	typical angina		1		**Max**	**4**			
3	nonangina		1						
4	asymptomaticangina		1		**Range**	**3**			
3	nonangina		1		**IQR**	**1**			
3	nonangina		1						
3	nonangina		1		**Mode**	**3**			
3	nonangina		1						
3	nonangina		1						
4	asymptomaticangina		1						

... | **Chest pain** | BP | Chol | BSugar | ECG | m Heart Beat | Exercise angina | OldPeak | Slope | heart Vessle

Figure 25Figure 24. Data pre processing (c)

Furthermore, the Chest pain attribute was categorical type data, where the nonanginal chest pain category is high.
The following screenshot is of the cleaned and categorized dataset.

1	Age	sex	CP	BP	Chol	B_sugar	ECG	H_Rate	Ex.Angine	Oldpeak	Slope	Vessele	Thalassemi	class
2	U80	Male	asymptomaticangina	High	High Rsk	High	Abnormal	Normal	Yes	High	Flat	Coloured	No	present
3	U70	Female	nonangina	Normal	High Rsk	Low	Normal	High	No	Low	Flat	Not Coloured	Fixed	absent
4	U60	Male	atypical angina	High	High Rsk	High	Abnormal	High	Yes	Low	Upsloping	Not Coloured	Reversible	present
5	U70	Male	nonangina	High	Normal	High	Normal	Normal	Yes	Low	Flat	Not Coloured	Fixed	absent
6	U80	Female	nonangina	Normal	Normal	High	Hipertrophy	Normal	Yes	Low	Upsloping	Not Coloured	No	absent
7	U70	Male	nonangina	Normal	Normal	High	Normal	High	No	Low	Upsloping	Not Coloured	No	absent
8	U60	Male	asymptomaticangina	High	High Rsk	High	Abnormal	High	Yes	Low	Flat	Coloured	Fixed	present
9	U60	Male	asymptomaticangina	Normal	High	High	Abnormal	High	Yes	Low	Flat	Coloured	Reversible	present
10	U70	Male	asymptomaticangina	High	High Rsk	High	Abnormal	High	Yes	High	Flat	Coloured	Reversible	present
11	U70	Female	asymptomaticangina	High	High Rsk	High	Hipertrophy	High	No	High	Flat	Coloured	Reversible	present
12	U60	Male	nonangina	Normal	Normal	Low	Abnormal	High	No	Low	Flat	Not Coloured	No	absent
13	U60	Male	nonangina	Normal	Normal	Low	Abnormal	Normal	Yes	Low	Upsloping	Not Coloured	No	absent
14	U50	Male	nonangina	Normal	High	Low	Abnormal	High	No	Low	Upsloping	Not Coloured	No	absent
15	U70	Male	typical angina	High	High	Low	Normal	High	No	High	Flat	Coloured	Fixed	present
16	U60	Female	nonangina	High	Normal	Low	Hipertrophy	High	No	Low	Upsloping	Not Coloured	No	absent
17	U80	Female	nonangina	Normal	Normal	Low	Normal	Normal	No	Low	Upsloping	Not Coloured	No	absent
18	U50	Male	asymptomaticangina	High	High Rsk	High	Abnormal	Normal	Yes	High	Flat	Coloured	Reversible	present
19	U60	Male	asymptomaticangina	High	High Rsk	High	Abnormal	High	Yes	High	Downsloping	Coloured	Reversible	present
20	U70	Male	typical angina	Normal	High	Low	Normal	High	No	High	Upsloping	Not Coloured	No	absent
21	U50	Male	nonangina	High	Normal	High	Normal	High	Yes	High	Upsloping	Not Coloured	Reversible	absent
22	U70	Male	asymptomaticangina	High	High	High	Abnormal	Normal	Yes	High	Flat	Coloured	Reversible	present
23	U50	Male	nonangina	Normal	High	Low	Normal	High	No	Low	Upsloping	Not Coloured	No	absent
24	U50	Male	nonangina	Normal	Normal	Low	Abnormal	High	No	Low	Upsloping	Not Coloured	No	absent
25	U50	Male	nonangina	Normal	Normal	Low	Normal	Normal	No	Low	Upsloping	Not Coloured	No	absent
26	U60	Female	nonangina	High	Normal	Low	Abnormal	High	Yes	Low	Upsloping	Coloured	No	absent
27	U50	Female	nonangina	High	Normal	High	Abnormal	Normal	No	Low	Upsloping	Not Coloured	Fixed	absent
28	U50	Female	asymptomaticangina	High	Normal	High	Abnormal	High	No	Low	Upsloping	Not Coloured	No	absent
29	U60	Female	nonangina	Normal	Normal	Low	Abnormal	High	No	Low	Upsloping	Not Coloured	No	absent
30	U60	Male	nonangina	High	High	High	Abnormal	High	Yes	High	Flat	Coloured	Reversible	present
31	U80	Female	nonangina	Normal	High	High	Abnormal	High	No	Low	Upsloping	Coloured	No	absent
32	U60	Male	nonangina	High	High	Low	Abnormal	High	No	Low	Upsloping	Coloured	Reversible	present
33	U70	Male	asymptomaticangina	Normal	High	Low	Abnormal	High	No	High	Upsloping	Not Coloured	Fixed	absent
34	U40	Female	nonangina	Normal	High	Low	Normal	High	No	Low	Upsloping	Not Coloured	No	absent
35	U60	Male	asymptomaticangina	High	High Rsk	High	Hipertrophy	High	Yes	High	Downsloping	Not Coloured	Reversible	present
36	U60	Male	typical angina	High	High Rsk	High	Abnormal	Normal	Yes	Low	Flat	Not Coloured	Reversible	present
37	U50	Male	asymptomaticangina	High	High Rsk	High	Abnormal	High	Yes	Low	Downsloping	Coloured	Reversible	present

Figure 26Figure 25. Dataset categorical

As a part of pre-processing the data, there were no attributes found that did not have a dependency on the prediction of disease. The final categorical dataset with 13 attributes, including class variables, was used in the Weka tool.

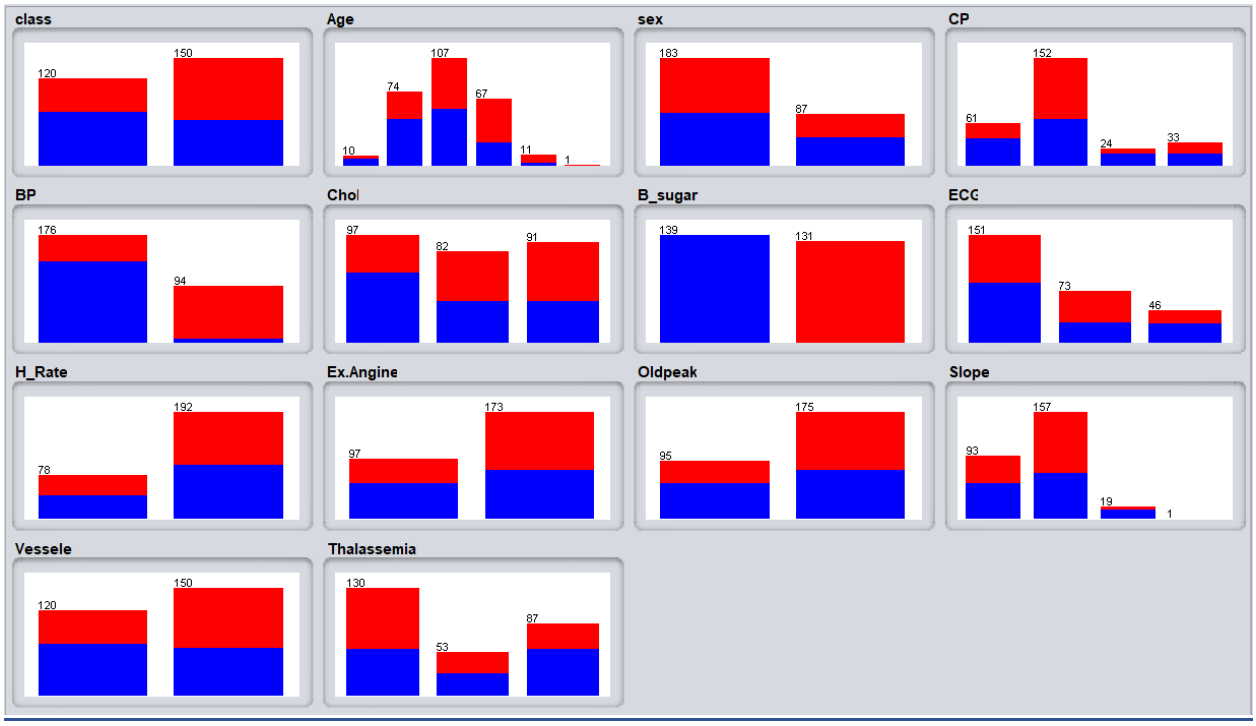


Figure 27Figure 26. Visualization of all attributes in Weka

Data Mining

By analyzing massive data sets by identifying patterns and linkages. Four steps comprise the data mining process.:

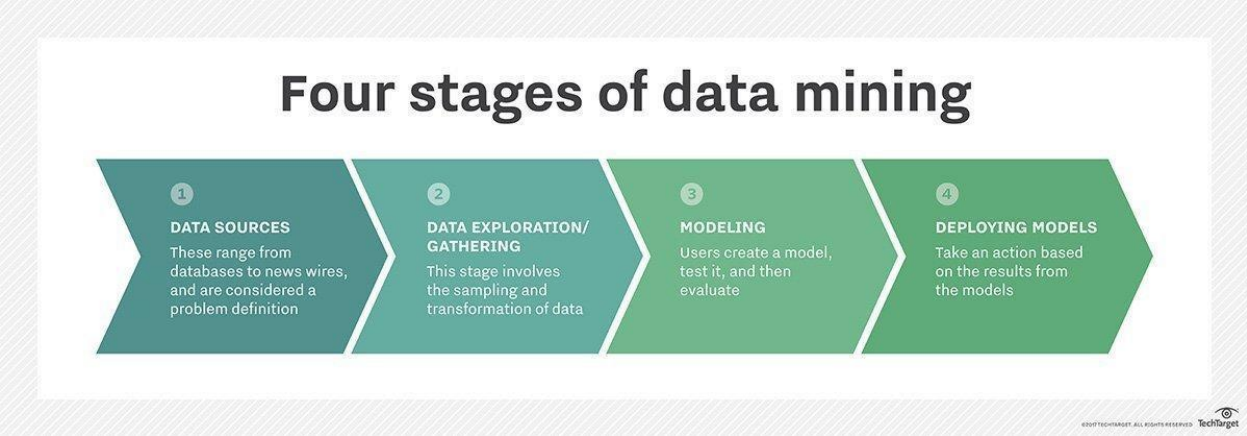


Figure 28Figure 27. data mining process

Out of various algorithm classifications, approaches were decided to visualize the data. (Craig, n.d.)

Data mining algorithm to analyse your preprocessed data set further.

After completing the preprocessing and transformation, the dataset was plugged into the Weka Tool for data mining.

1) The transformed data set was open in the preprocess tab by following Explorer > open file> Recent > New folder>heart disease record.csv

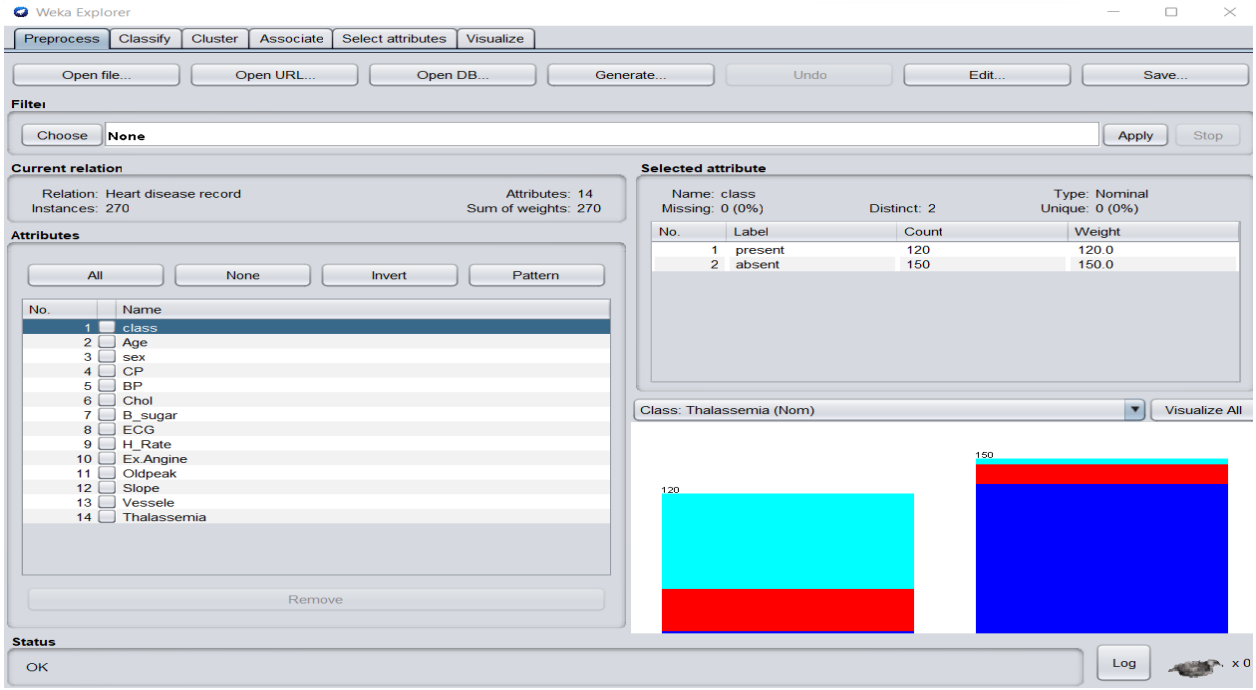


Figure 29Figure 28. data mining pre processing in weka

2) Open the classify tab to apply the decision tree-j48 algorithm

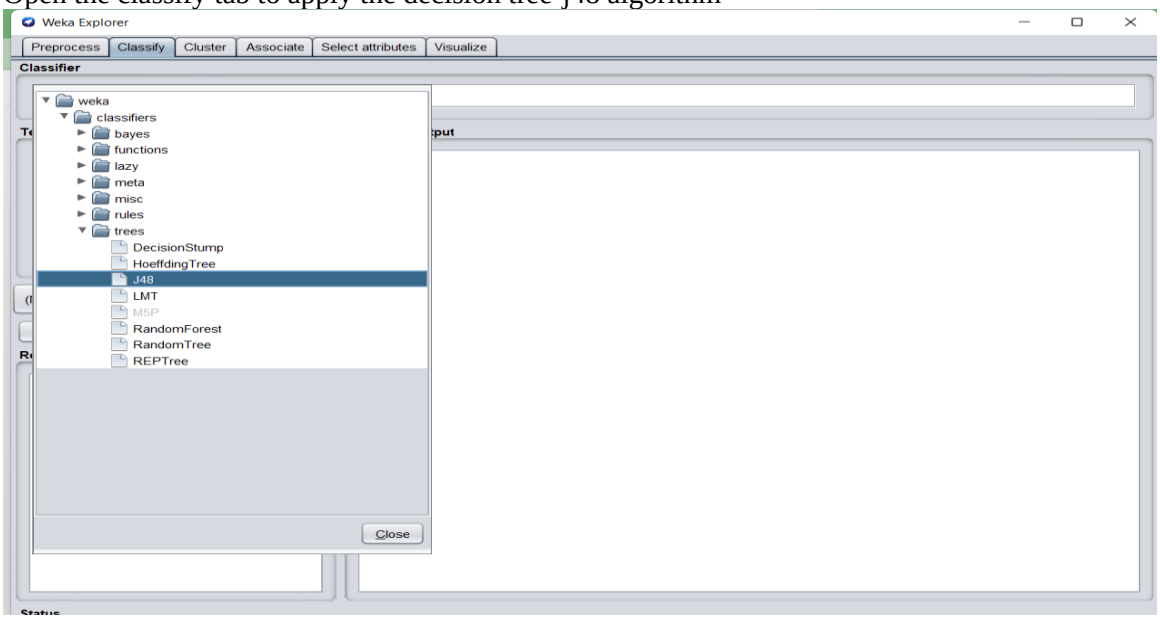


Figure 30Figure 29. classifying j48 in weka

3) On clicking the start button, the algorithm was run on the dataset, and the following output was generated

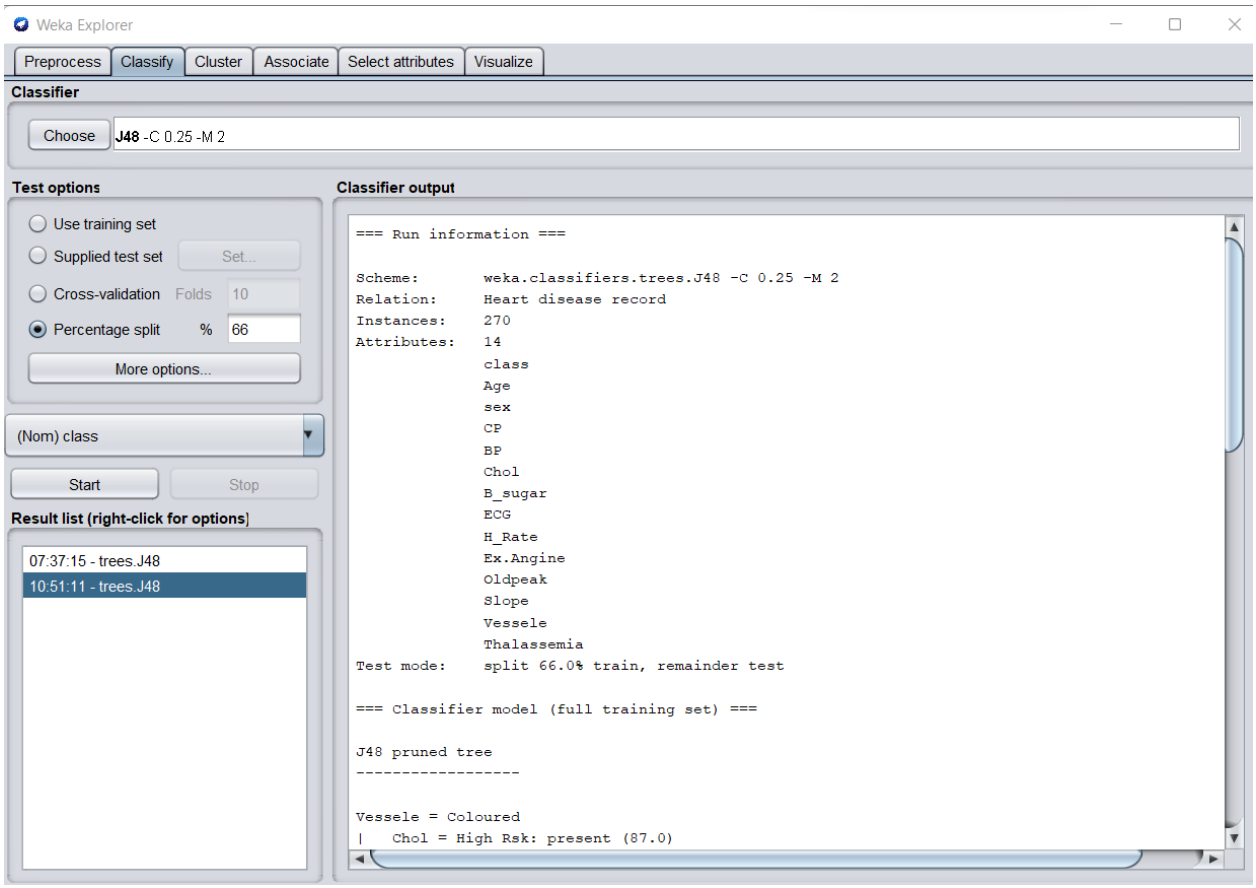


Figure 31Figure 30. classifier output in weka

The above findings show that the model building time was 2 seconds, 0.1-second split test. 87 instances were with 92.3% accuracy, and 7 were incorrectly classified.

Implementation of the data mining algorithm.

```
=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.918    0.070    0.938      0.918    0.928      0.848    0.964     0.964     present
                 0.930    0.082    0.909      0.930    0.920      0.848    0.964     0.943     absent
Weighted Avg.    0.924    0.075    0.924      0.924    0.924      0.848    0.964     0.954

=== Confusion Matrix ===

  a  b   <-- classified as
 45  4 |  a = present
  3 40 |  b = absent
```

Figure 32Figure 31. confusion Metrix in weka

The confusion matrix According to the findings, 45 instances were accurately identified as placed, whereas 4 instances were wrongly classified as not placed. 40 instances that are actually not placed were correctly classified as not placed,
whereas 3 instances that were actually not placed were incorrectly classified as placed.

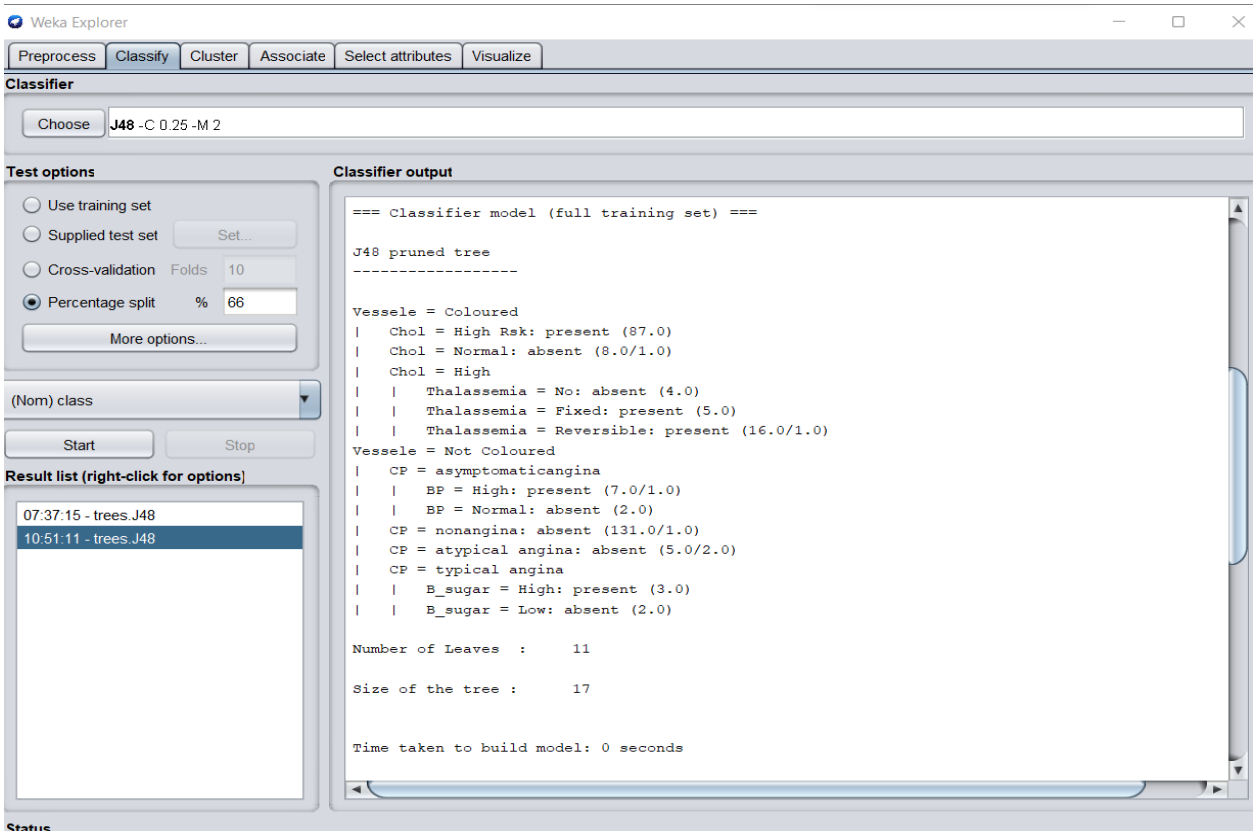


Figure 33Figure 32. rules in weka

The tree has 11 leaves (the pure nodes which are either placed/not placed) with a tree size of 17. The tree was also pruned to improve the prediction accuracy & reduce the problems caused due to overfitting.

Interpreting the results

The tree, when visualized:

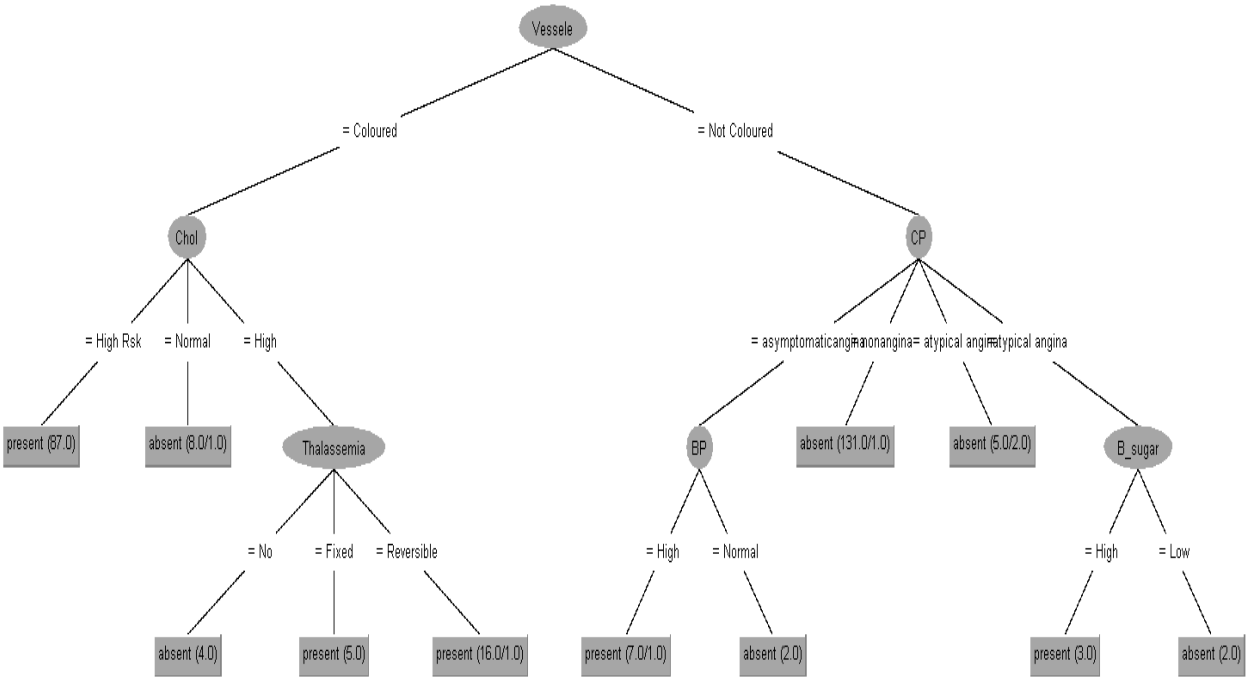


Figure 34Figure 33. Decision tree in weka

The rules derived from the tree are as follows:

1. If the patient's vessels are coloured and Cholesterol is at high risk, then heart disease is present.
2. If the patient's vessels are coloured, and Cholesterol is normal, then heart disease is absent.
3. If the patient's vessels are coloured, Cholesterol is high, and Thalassemia is No, then heart disease is absent.
4. If the patient's vessels are coloured, Cholesterol is high, and Thalassemia is fixed, then heart disease is present.
5. If the patient's vessels are coloured, Cholesterol is high, and Thalassemia is reversible, then heart disease is present.
6. If the patient's vessels are not coloured and chest pain with atypical angina, then heart disease is absent.

7. If the patient's vessels are not coloured, and chest pain is nonanginal, then heart disease is absent.
8. If the patient's vessels are not coloured and chest pain with nonangina, then heart disease is absent.
9. If the patient's vessels are not coloured and chest pain with nonangina, then heart disease is absent.
10. If the patient's vessels are not coloured, chest pain with asymptomatic angina, and B.P. (resting blood pressure) is high; then heart disease is present.
11. If the patient's vessels are not coloured, chest pain with asymptomatic angina, and B.P. (resting blood pressure) is Normal; then heart disease is absent.
12. If the patient's vessels are not coloured, chest pain with typical angina, and B_sugar(fasting blood sugar) is high; then there is a possibility of heart disease.
13. If the patient's vessels are not coloured, chest pain with typical angina, and B_sugar(fasting blood sugar) is Low; then heart disease is absent.

Hence the data mining using the decision tree (148) algorithm was successfully implemented using Weka Tool.

Data Ethics

Growing concern about 'Data Ethics' has arisen due to an increase in data usage for analytical purposes. Moreover, data scientists or analysts do so without following specific ethical guidelines when dealing with personal data.

Data Ethics is a subset of ethics that deals with concerns related to data. Data capture, sharing, processing, curation, and programmatic hacking all need adherence to ethical principles. (*What Is Data Ethics? | Cognizant*, n.d.)

Analyzing and exploiting corporate data is fraught with ethical dilemmas. Among the essential ideas are:

- A consent form is a crucial problem with data usage, where the participant is required to offer comprehensive approval and is conscious of how their data is being handled and will be treated in the future.
- The person should know the goal, nature, possible advantages, hazards, etc. This concept is often broken, resulting in legal action.
- Customers are ignorant of how their data is processed via a huge chain of analytical tools since organizations typically do not identify the objective of data utilization.
- Data processing and analysis sometimes cause unfair biases that reveal problems like sexism and racism. Individual data are now subject to machine learning biases and human prejudices. Disparity and maltreatment are unethical when caused by prejudice.
- Organizations should not distribute data carelessly. In many circumstances, data is freely shared. Unauthorized third parties will share this data with other organizations. These data will be misused, producing problems for both consumers and businesses. That way, exploitation may be avoided. The organizations must protect the data's anonymity, confidentiality, and privacy.
- Most organizations do not protect or do not offer security for data acquired for data analytics, which allows hackers and cybercriminals to exploit the data providers' worries.
- Al-attacks are when Artificial Intelligence gains unauthorized access to data and exploits it. It might be DOS, evasion, or data poisoning.
- Client analysis, where gathered data generates a customer profile, impacting qualities such as the financial ability. However, inaccurate consumer profiling may lead to inaccurate statements about customers, which is immoral.

These are serious ethical dilemmas. In academics, there are additional ethical and privacy concerns. Health organizations gather all clinical data. Health data analysts expect these organizations to preserve their privacy as data suppliers.
Many universities monitor WIFI networks and phone sensors to track student performance and virtual behaviour; however, this violates student privacy and is unethical.

The GDPR (General Data Protection Regulation) safeguards the acquired data's security, privacy, confidentiality, and anonymity. (GDPR. Eu, n.d.) The GDPR offers individuals control over their data and enables enterprises to follow a standard ethical code.

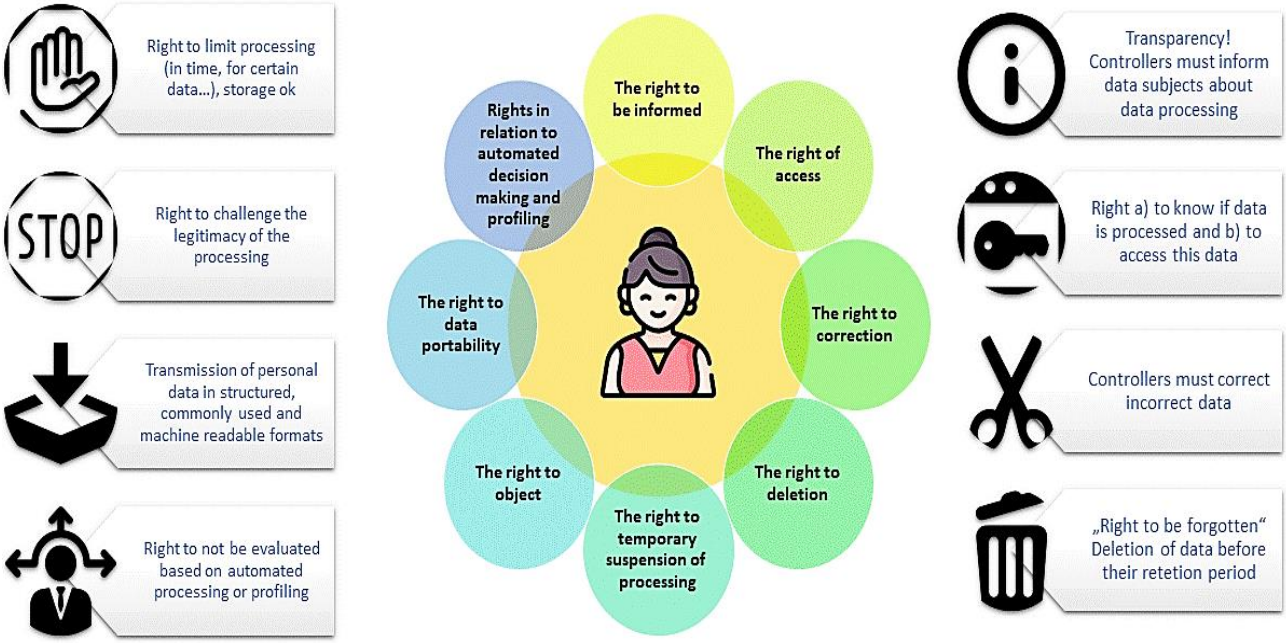


Figure 35Figure 34. Data Ethics

Conclusion

A discussion of the overall visualization results

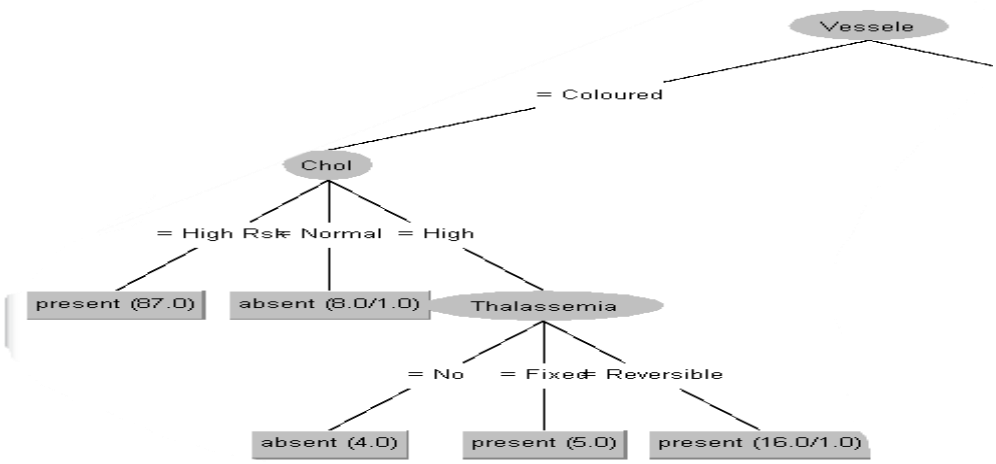
On applying this dataset for data analysis in Tableau and data mining in Weka, similar
Moreover, findings can be found.
Heart disease is difficult to diagnose due to contributing risk factors such as diabetes, high blood pressure, excessive Cholesterol, irregular pulse rate, and others.
However, as per the data mining performed on Weka, if the patients have coloured vessels, high bp, high Cholesterol, and fixed thalassemia are diagnosed with heart disease. Those diagnosed with low sugar, normal B.P., Normal Cholesterol, and atypical angina chest pain are more likely to be absent from heart disease.

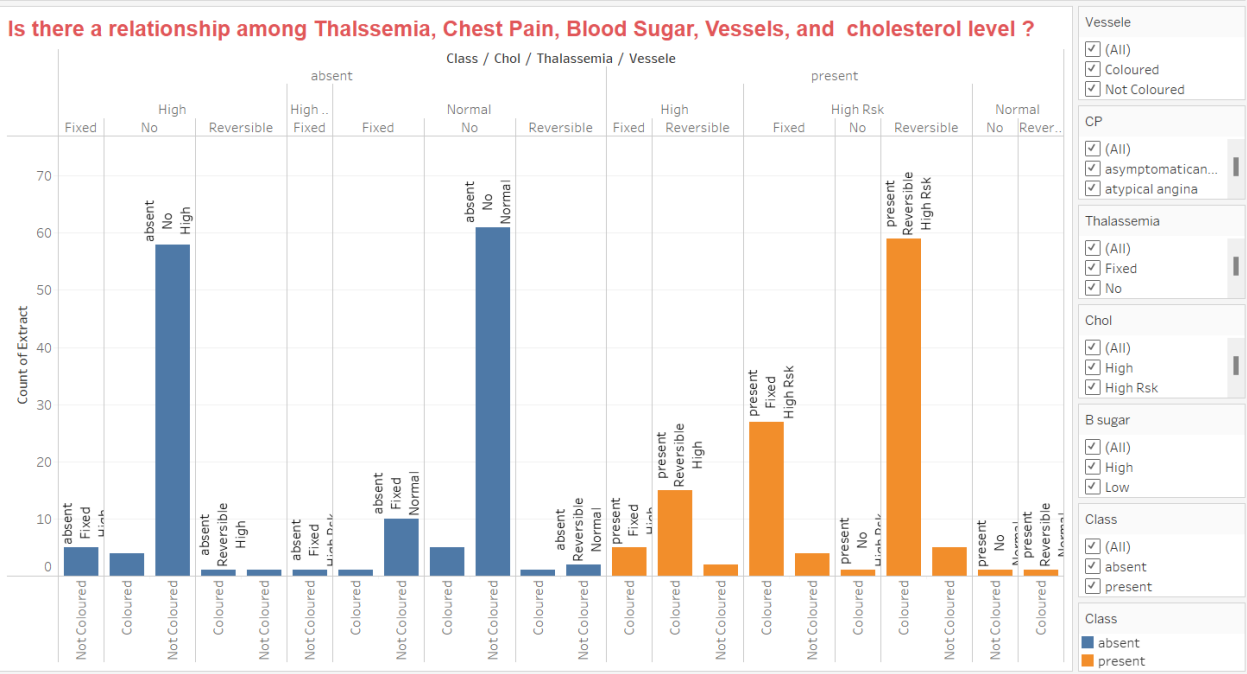
Trends and patterns

In terms of Business Intelligence, people who wish to avoid heart disease should concentrate on regular checkups, good food, and exercise.
According to some studies, heart disease is the leading cause of death for both men and women. Every year, cardiac attacks occur. 525,000 new, 210,000 repeat. Every year, cardiac attacks occur. 525,000 new, 210,000 repeat.
Heart disease is a major concern. However, diabetes, high blood pressure, high Cholesterol, irregular pulse rate, and other risk factors make identification difficult.
These limits led scientists to use current methods like data mining and machine learning to forecast illness.
Machine learning (ML) uses vast volumes of data from the medical industry to create predictions. The data visualizations and data mining also helped people concentrate on their health. Results in Weka and Tableau are essential.

A discussion of the data mining results

This data is relevant to this investigation. Even some patients with High Blood sugar are free from heart disease and can live longer.





According to the Weka norm, if a patient's vessels are coloured (blocked) and Cholesterol normal, then heart disease is not found. Similarly, if Thalassemia is identified as 'No', the disease is absent even though Cholesterol is high.
Similar results are also found in tableau analysis; the patients with high- or high-risk Cholesterol are 59%, and for those who have No or reversible Thalassemia with normal or high-risk Cholesterol, coloured vessels are only 1%—and diagnosed with heart disease.

Another analysis shows that though the vessels are not coloured, Chest pain is asymptomatic and typical, B.P. high, and B_sugar high, then heart disease is present. Therefore, we cannot rely on only a few factors.

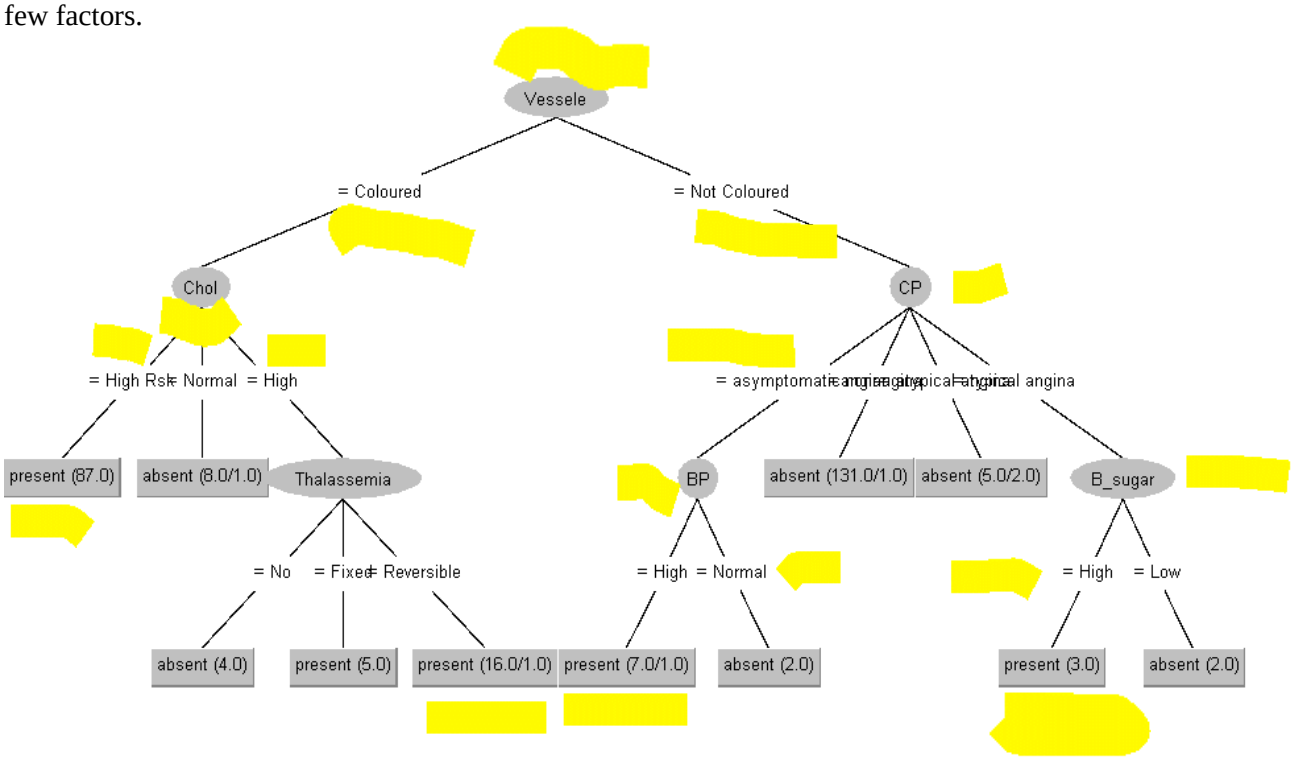
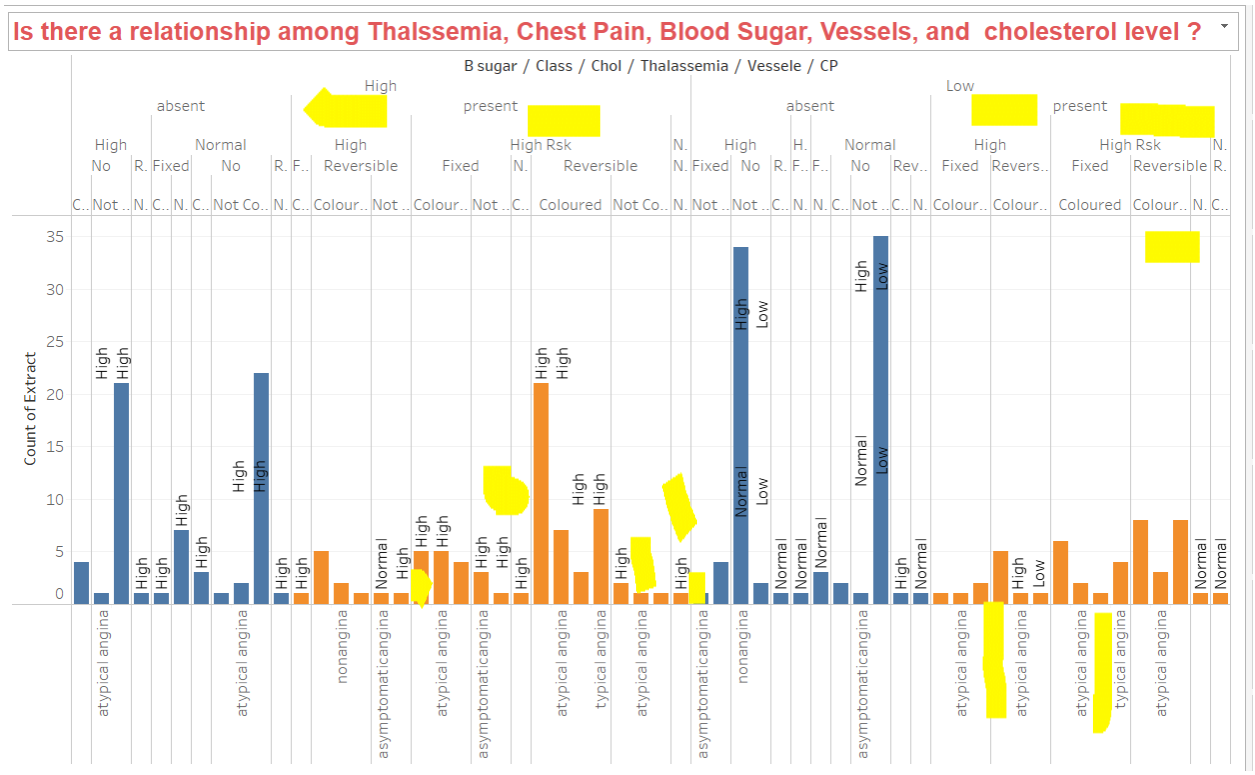


Tableau also visualized a similar result as Weka, which has already been discussed in the data mining section.



Recommendations

People who wish to lower their risk of heart disease should receive regular blood pressure, exercise stress tests, Cholesterol, and blood sugar checks. Nowadays, convenient test machines have emerged. So they can self-check. Also, machine learning and A.I. in healthcare enable extensive symptom analysis.

Analyzing precise physical and biochemical data makes it possible to foresee heart illness. Aside from a few studies, there is no broadly accepted machine learning CVD diagnostic approach.

Data mining was successful based on the initial findings, as seen on Tableau. The classification accuracy of this dataset (92.3%) is good. This dataset's analysis and mining results reflect similar tendencies. This was a success.

A discussion of the business intelligence that can be obtained from these results.

- To decrease turnover, clients might be charged solely based on consumption.
- Another alternative is to send out frequent surveys to customers to check their happiness.

- Clients who regularly use Orange Telecom services might also be offered discounts to encourage them to remain and utilize them more often.

References

Decision Tree Algorithm Examples in Data Mining. (n.d.). Retrieved April 15, 2022, from https://www.softwaretestinghelp.com/decision-tree-algorithm-examples-data-mining/

What Is Data Ethics? | Cognizant. (n.d.). Retrieved April 15, 2022, from https://www.cognizant.com/us/en/glossary/data-ethics?msclkid=4773e391bcdf11ecb51d8c023fe69c52

What is Data Mining? (n.d.). Retrieved April 15, 2022, from https://www.techtarget.com/searchbusinessanalytics/definition/data-mining

What is GDPR, the E.U.'s new data protection law? - GDPR.eu. (n.d.). Retrieved April 15, 2022, from https://gdpr.eu/what-is-gdpr/?msclkid=a56ed010bcdf11ec952e7dace68e595b